# Frequentist and Bayesian Analysis of the Matter–Antimatter Asymmetry in $B^{\pm} \to K^{\pm}K^{+}K^{-}$ Decays at LHCb

**Final project for the *Statistics and Data Analysis* course**
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We measure the raw charge asymmetry $A_{\rm raw} = [N(B^-) - N(B^+)]/[N(B^-) + N(B^+)]$ of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays in the 2011 LHCb Open Data record with an extended unbinned maximum-likelihood fit and with a Bayesian analysis of the same likelihood. On 19,731 candidates we find $\boldsymbol{A_{\rm raw} = -0.0441 \pm 0.0085\,(\rm stat) \pm 0.0072\,(\rm syst)}$ (Frequentist) and $\boldsymbol{-0.0443 \pm 0.0085 \pm 0.0070}$ (Bayesian), **5.2 standard deviations** from zero. The methods agree in value, uncertainty and coverage, as verified on toy-MC experiments, and the dominant systematic uncertainty is the *particle-identification requirement*. In the Dalitz region of the published analysis we find $\boldsymbol{-0.234 \pm 0.025}$, reproducing $-0.239 \pm 0.020$. A machine-learning selection trained on weak labels leaves $A_{\rm raw}$ unchanged within its statistical spread. $A_{\rm raw}$ is not corrected for production and detection asymmetries and is not a measurement of $A_{CP}$.

## I. INTRODUCTION

The observable of this work is the raw charge asymmetry of the signal yields of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays,

$$A_{\rm raw} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}, \tag{1}$$

which in three-body charmless $B$ decays can be studied globally and locally over the Dalitz plane [1, 2]. It differs from the $CP$ asymmetry $A_{CP}$ by the production asymmetry $A_P$ of $B^{\pm}$ in $pp$ collisions and by the detection asymmetry $A_D$ of the final state [1],

$$A_{CP} \approx A_{\rm raw} - A_P - A_D. \tag{2}$$

Neither correction is applied here, so $A_{\rm raw}$ is *not* a measurement of $A_{CP}$; the corrections of the published analysis are discussed in Sec. X. We use the LHCb Open Data record for this decay [3, 4], a subset of the 2011 data on which the first evidence of an inclusive $CP$ asymmetry was published [5], to ask: *to what extent do Frequentist likelihood fits and Bayesian inference produce consistent, well-controlled estimates of $A_{\rm raw}$?* The aim is a methodological comparison of the two frameworks on the same data and on the same likelihood, so that any difference comes from the inference and not from the model. Secs. II–III describe the data and the common model, Secs. IV–V the two fits, and Sec. VI the validation of the estimator and of the intervals, Sec. VII the machine-learning analysis, Sec. VIII the Dalitz plane, Sec. IX the systematic study and the result, and Sec. X the comparison. The main results are in Table I; the symbols are listed in Appendix J.

## II. DATA

*In short: the record, the common preselection (*23,895 *candidates) and the fit window (*19,731 *candidates); the spectrum peaks at* 5284 *MeV for both charges and both polarities, and the counting asymmetry is* $-0.043 \pm 0.007$.

### A. Dataset, Preselection and Mass

The dataset is the CERN Open Data record 4900 [3]: $B^{\pm} \to h^{\pm}h^{+}h^{-}$ candidates collected by LHCb in 2011 at $\sqrt{s} = 7\,\text{TeV}$, already through trigger and offline selection, split by magnet polarity (8,556,118 candidates), and a detector-free phase-space simulation of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ (50,000 decays, no asymmetry) used only to test the local-asymmetry procedure (Sec. VIII). The same preselection is used in all analyses,

$$\texttt{H*_ProbPi} < 0.5 \;\wedge\; \texttt{H*_ProbK} > 0.5 \;\wedge\; \neg\texttt{H*_isMuon},$$

for each of the three tracks, and retains 23,895 candidates (0.28%) (Appendix A.1). $H1$ is always the kaon of charge opposite to the $B$ and $H2$, $H3$ are the same-sign pair, so the Dalitz variables are the two opposite-sign invariant masses $m^2(H1, H2)$ and $m^2(H1, H3)$. The three tracks are assigned the charged-kaon mass and their four-momenta are summed to reconstruct $m(KKK)$. The **nominal fit window**, common to the two analyses, is $[5100, 5600]$ MeV and contains 19,731 candidates (11,636 magnet down, 8095 magnet up). Below ∼5150 MeV the background rises, which a single exponential cannot describe; the window includes the start of this rise, described by a second exponential in the background model (Secs. IV–V).

### B. Data Exploration

Figure I shows the spectrum in the fit window. The peak at $m \approx 5284\,\text{MeV}$ over a falling combinatorial background has the same shape for the two charges and the two polarities, which allows a mass and a resolution

| Method | Sample | $A_{\rm raw}$ | Statistical | Systematic | Section |
|---|---|---|---|---|---|
| Counting (no fit) | fit window, 19,731 | $-0.0432$ | 0.0071 | – | II B |
| Frequentist fit | fit window, 19,731 | $-0.0441$ | 0.0085 | 0.0072 | IV, IX |
| Bayesian fit | fit window, 19,731 | $-0.0443$ | 0.0085 | 0.0070 | V, IX |
| Bayesian fit, $D^0$ veto | 18,744 | $-0.0496$ | 0.0088 | – | App. H.1 |
| Bayesian fit, ML selection | score $> 0.5$, 15,709 | $-0.0451$ | 0.0087 | – | VII C |
| Frequentist fit, ML selection | score $> 0.5$, 15,709 | $-0.0447$ | 0.0086 | – | VII C |
| Bayesian fit, Dalitz region of Ref. [5] | 2146 | $-0.234$ | 0.025 | – | VIII |
| Published analysis (same data) [5] | different selection and model | $-0.060$ | 0.007 | – | – |

TABLE I. Summary of the raw asymmetry $A_{\rm raw}$ (not corrected for production and detection asymmetries). The Bayesian value is the posterior median and its standard deviation; the systematic uncertainty is given only for the nominal fits (Sec. IX). The published $A_{\rm raw}$ comes from a different selection and a model with additional background components, so it is not directly comparable with the first rows.

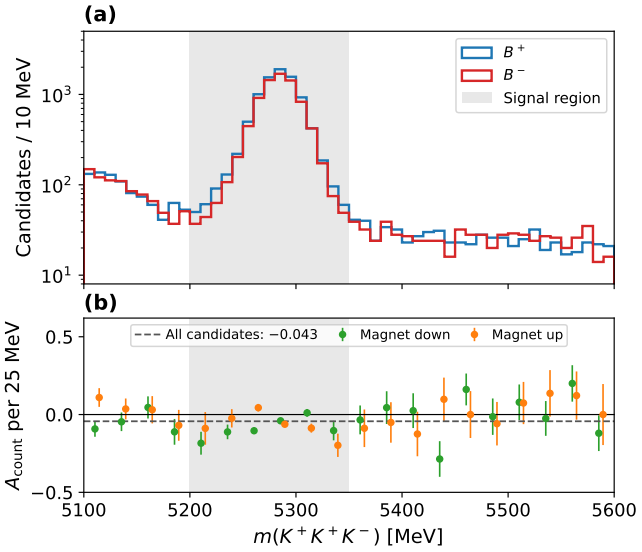



FIG. I. (a) Reconstructed $B \to KKK$ invariant mass in the nominal fit window $[5100, 5600]$ MeV for $B^+$ and $B^-$, logarithmic scale; the signal region $[5200, 5350]$ MeV is shaded. (b) Counting asymmetry in 25 MeV bins of the mass for the two magnet polarities, and for all candidates (dashed); the two polarities agree within their uncertainties and the asymmetry is similar in the peak and in the sidebands (Appendix D.8, Fig. XXII).

common to $B^+$ and $B^-$ in the model (Sec. III). The *counting asymmetry* of all candidates, signal and background together, is $A_{\rm count} = (N^- - N^+)/(N^- + N^+) = -0.043 \pm 0.007$ (binomial uncertainty); the fits of Secs. IV and V refine it by separating signal and background.

### C. Additional Results

*Selection.* The kaon-identification requirement on the three tracks is by far the most selective step of the preselection, and only 42 candidates of the fit window (0.2%) share a track with another candidate; all are kept (Appendix A.2, Table II).

## III. MASS MODEL AND LIKELIHOOD

*In short: one extended likelihood shared by the two analyses: a Gaussian signal with common mean and width, a double-exponential background for each charge and $A$ as a parameter.* The mass spectrum of each charge sample is described by a signal-plus-background mixture on the nominal fit window $[5100, 5600]$ MeV (Sec. II A),

$$f^{\pm}(m) = N_s^{\pm}\, S(m; \mu, \sigma) + N_b^{\pm}\, B^{\pm}(m), \qquad (3)$$

where $S$ is a Gaussian truncated to the window, with mean $\mu$ and width $\sigma$ shared by the two charges, and the background is a double exponential with a shape of its own for each charge,

$$B^{\pm}(m) = f_{\pm}\, E(m; \lambda_1^{\pm}) + (1 - f_{\pm})\, E(m; \lambda_2^{\pm}),$$

with $E(m; \lambda) \propto e^{-\lambda m}$ a truncated exponential normalized on the window. The steep component ($\lambda_1$) describes the rise of the background towards the lower edge of the window, the slow one ($\lambda_2$) the combinatorial background under and above the peak. The ordering $\lambda_1 > \lambda_2$ removes the label-switching symmetry of the mixture (exchanging the two exponentials would give two identical modes). Signal and background yields are split between the charges as

$$N_s^{\pm} = \tfrac{N_s}{2}(1 \mp A), \qquad N_b^{\pm} = \tfrac{N_b}{2}(1 \mp A_b),$$

so that $A = (N_s^- - N_s^+)/(N_s^- + N_s^+)$, the raw asymmetry $A_{\rm raw}$ of Eq. (1), is itself a parameter of the fit, the *parameter of interest* (POI, all the others being nuisance parameters), and the background asymmetry $A_b$ is free. The parameter vector has twelve components, $\theta = (N_s, N_b, A, A_b, \mu, \sigma, \lambda_1^{\pm}, \lambda_2^{\pm}, f_{\pm})$. Given the data $D = \{(m_i, q_i)\}_{i=1}^{N}$ (mass and charge of each candidate), the extended unbinned likelihood of the two charge sam-

ples is

$$\mathcal{L}(D \mid \theta) \propto e^{-(N_s+N_b)} \prod_{i:\, q_i=+1} \left[N_s^+ S(m_i)+N_b^+ B^+(m_i)\right] \times \prod_{i:\, q_i=-1} \left[N_s^- S(m_i) + N_b^- B^-(m_i)\right], \quad (4)$$

where $S(m_i) \equiv S(m_i;\mu,\sigma)$. The likelihood is *extended* (the yields are parameters, hence the Poisson term (Appendix B.1)); it is the common ground of the two mass analyses: Sec. IV maximizes it and Sec. V multiplies it by a prior. Since both use this model unchanged, any difference between their results can only come from *the inference*.

### A. Additional Results

*Minimal model.* A six-parameter model on a narrower window gives $A_{\rm raw} = -0.0497$ (68% Confidence Interval $[-0.058, -0.041]$), compatible with the nominal result within its uncertainty.

## IV. FREQUENTIST ANALYSIS

*In short: the likelihood maximized for the two charges together gives* $N_s = 15{,}328 \pm 133$ *and* $\hat{A} = -0.0441$ *with Minos intervals;* $A = 0$ *is rejected at* $5.2\sigma$ *and the interval has nominal coverage in* 1000 *bootstrap replicas.* The Frequentist analysis estimates $A_{\rm raw}$ by maximizing the extended unbinned likelihood of Eq. (4) with `iminuit` [6]; the uncertainty and the test of $A = 0$ come from the profile likelihood, the other parameters being nuisance parameters (Secs. IV A–IV B).

### A. Mass Model and Fit Strategy

The common mass model of Sec. III (Eq. (3)) was fitted on $[5100, 5600]$ MeV: a Gaussian signal with mean $\mu$ and width $\sigma$ shared by the two charges and a double-exponential background with shape and yield of its own for each charge, twelve parameters as in the Bayesian fit, through `iminuit.cost.ExtendedUnbinnedNLL`; the minimization started from an unbinned fit of the sidebands (the window without $[5200, 5350]$ MeV). The asymmetry was estimated in two ways: in the *simultaneous* fit the two charges share $N_s^\pm = N_s(1 \mp A)/2$ and $A$ is a parameter of the likelihood; in the *separate* fits each charge was fitted alone (seven free parameters each) and $A_{\rm raw} = (N_s^- - N_s^+)/(N_s^- + N_s^+)$ was computed from the two yields. They give $-0.0441 \pm 0.0085$ and $-0.0442 \pm 0.0087$: sharing $\mu$ and $\sigma$ between a particle and its antiparticle is adequate, and the simultaneous fit, which gives the profile-likelihood interval directly, is the **nominal** one. It gives $N_s = 15{,}328 \pm 133$ and $N_b = 4403 \pm 82$, against the Bayesian posterior medians $15{,}328 \pm 133$ and $4404 \pm 82$ (Sec. V C). The nominal fit has $\chi^2/\mathrm{ndof} = 160/88$ ($p = 4.8 \cdot 10^{-6}$) in $2 \times 50$ bins of 10 MeV; a single exponential is rejected (294/92, $p = 5.7 \cdot 10^{-23}$) (Appendix E.3, Fig. XXVI) because the background rises towards the lower edge of the window (Sec. II A). The fit of the two charges is shown in Fig. II.

### B. Profile Likelihood and Confidence Interval

The maximum-likelihood estimate is $\boldsymbol{\hat{A} = -0.0441}$ (Hesse uncertainty 0.0085), with `MINOS` [7] intervals 68% CI $[-0.0526, -0.0356]$ and 95% CI $[-0.0607, -0.0275]$; the uncertainties are statistical only (the systematic one is in Sec. IX). The *profile likelihood ratio* $q(A) = -2\log[L(A,\hat{\hat{\theta}})/L(\hat{A},\hat{\theta})]$, with the other eleven parameters re-minimized at each $A$ (`mnprofile`), is shown in Fig. III; by Wilks' theorem [8] it follows a $\chi^2$ distribution with one degree of freedom at the true value of $A$, so the intervals are the sets $q \le 1$ and $q \le 3.84$. For the null hypothesis, $q(0) \simeq 27$, $p = 2.05 \times 10^{-7}$, $\boldsymbol{5.20\sigma}$ (Appendix C.1): no asymmetry is rejected, the counterpart of the Savage-Dickey test of Sec. V D ($A = 0$ at 5.2 posterior standard deviations); the $p$-value is not the probability of $H_0$ and the Bayes factor depends on the prior width, but the conclusion is the same. The result agrees with the Bayesian one (Sec. V C: $-0.0443$, 68% CI $[-0.0528, -0.0358]$), as expected with flat priors and $\sim 2 \times 10^4$ events.

### C. Coverage of the Wilks Intervals

Wilks' theorem is asymptotic, so the coverage was checked on the data with 1000 bootstrap replicas of the 19,731 candidates, refitting the nominal model on each (non-converged fits discarded); with $A_{\rm true} = -0.0441$, the 68% interval of a replica contains it if and only if $q(A_{\rm true}) \le 1$. The fraction is $\boldsymbol{68.4 \pm 1.5\%}$ against 68.3% nominal (Fig. IV). It agrees with the toy-MC check of the Bayesian interval ($65.0 \pm 3.3\%$, 200 experiments) (Appendix D.10, Fig. XXIV); the two checks are not equivalent, since the Bayesian pseudo-datasets come from the fitted model (a closure test) while the replicas are resampled from the data and carry the features that the model does not describe, such as the signal tails (Appendix D.5, Table VII, Figs. XXI and XX).

### D. Additional Results

*Neyman belt construction and momentum.* Inverting the test of $A_{\rm true}$ with toy-MC experiments gives the same interval as the likelihood ratio, and $A_{\rm raw}$ does not depend on the momentum of the $B$ (Appendix C.2, Figs. XV and XVI, Table III). *Uncertainties of the twelve parameters.* The standard deviations from the Hesse matrix,

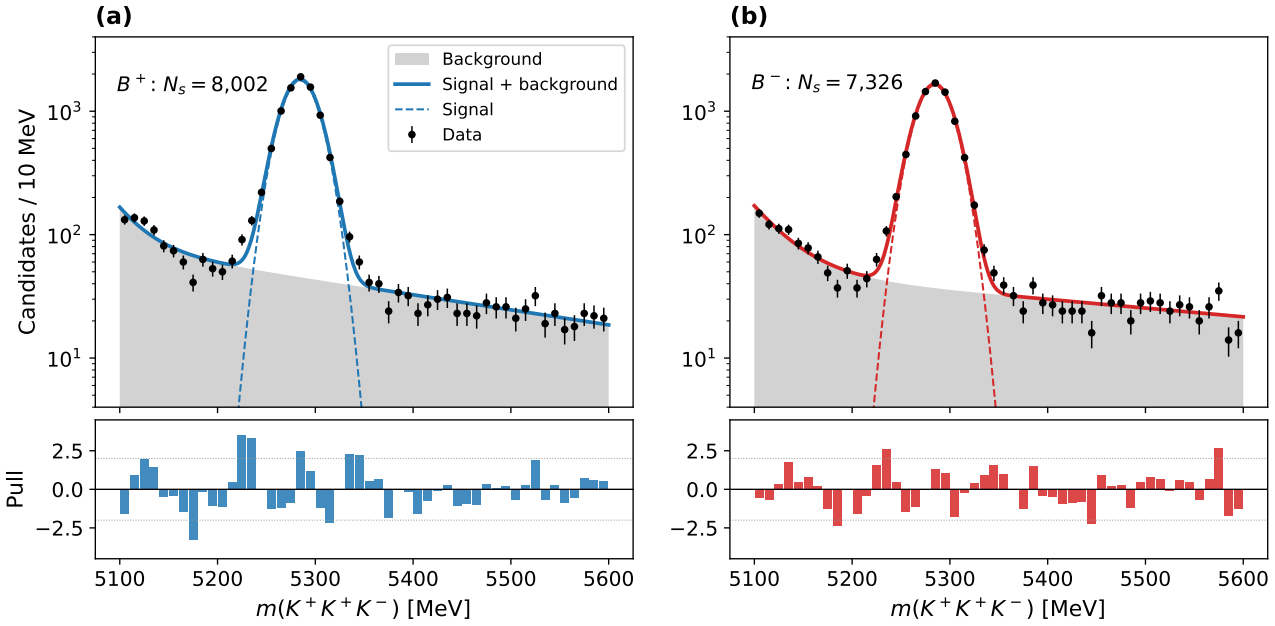



FIG. II. Simultaneous fit of (a) $B^+$ and (b) $B^-$ in 10 MeV bins, logarithmic scale: data, signal (dashed), double-exponential background (grey) and their sum; below, the pulls $(n_i - \mu_i)/\sqrt{\mu_i}$ of the fit, with the $\pm 2$ lines. The fitted signal yields are 8002 and 7326.

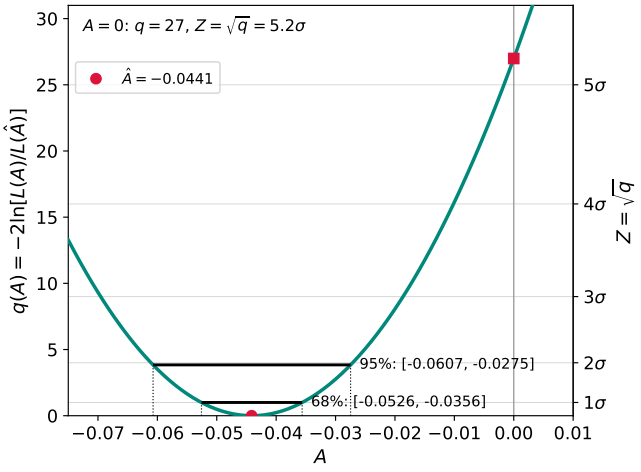



FIG. III. Profile likelihood ratio $q(A)$ of the simultaneous fit, with the other eleven parameters re-minimized at each $A$ (right axis: $Z = \sqrt{q}$). The horizontal segments are the 68% and 95% Minos intervals ($q \leq 1$, $q \leq 3.84$); at $A = 0$, $q \simeq 27$, i.e. $5.2\sigma$.

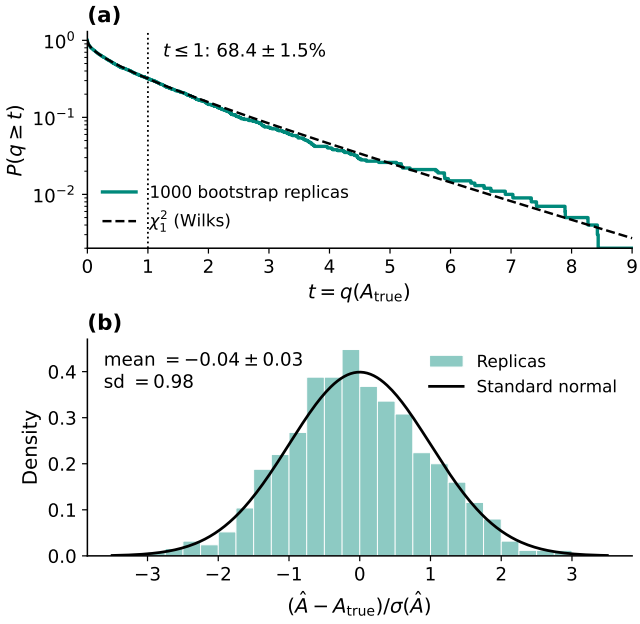



FIG. IV. Coverage check with 1000 bootstrap replicas of the data. (a) Survival function of $t = q(A_{\text{true}})$ against the $\chi^2_1$ distribution expected from Wilks' theorem; the dotted line is the 68.3% limit ($t = 1$). (b) Pulls $(\hat{A} - A_{\text{true}})/\sigma(\hat{A})$ of the replicas against a standard normal.

the bootstrap of the data and the posterior agree for $A$, $A_b$, $\mu$ and $\sigma$; the bootstrap differs for the two yields, because it keeps the total number of candidates fixed, and for the weakly constrained steep slope $\lambda_1^+$ (Appendix C.3, Table IV, Fig. XVII).

## V. BAYESIAN ANALYSIS

*In short: the same likelihood with uniform priors, sampled with MCMC: $A_{\mathrm{raw}} = -0.0443$ (68% interval $[-0.0528, -0.0358]$) with converged chains; the Bayes factor of $A = 0$, $BF_{10} \approx 9 \times 10^3$, depends on the prior width.* The Bayesian analysis takes the likelihood of Sec. III as it is and adds only a prior and a sampler; it changes the question asked of the data, not the model.

### A. Priors and Posterior

Bayes' theorem combines the likelihood of Eq. (4) (Sec. III) with a prior $\pi(\theta)$ into the posterior

$$p(\theta \mid D) = \frac{\mathcal{L}(D \mid \theta)\, \pi(\theta)}{p(D)}, \tag{5}$$

The priors are uniform, with independent ranges except for the ordering of the background slopes: $N_s, N_b \in (0, 5N)$ with $N = 19{,}731$, $A, A_b \in (-1, 1)$, $\mu \in (5260, 5310)$ MeV, $\sigma \in (3, 40)$ MeV, $\lambda_1^{\pm} \in (10^{-5}, 0.1)$ MeV$^{-1}$, $\lambda_2^{\pm} \in (10^{-5}, \lambda_1^{\pm})$, $f_{\pm} \in (0, 1)$. The prior density is therefore a constant on the allowed region $\Theta$,

$$\pi(\theta) \propto \mathbf{1}\left[\theta \in \Theta\right], \tag{6}$$

and inside it the posterior is proportional to the likelihood. A flat prior is not invariant under reparametrization, but with $\sim 2 \times 10^4$ events the posterior is dominated by the likelihood, as a prior-sensitivity check (Appendix D.1, Table V) confirms for $A$. The evidence $p(D)$ is a constant: the MCMC samples $\mathcal{L}\,\pi$ without it, and the Savage-Dickey ratio of Sec. V D avoids it as well; the marginal posterior of $A$ is the histogram of the $A$ component of the chain.

### B. MCMC Sampling and Convergence Diagnostics

The posterior was sampled with the affine-invariant ensemble sampler of Goodman and Weare [9] as implemented in `emcee` [10] (Appendix D.2): 32 walkers started around the maximum of the posterior, 25,000 steps with the first 1/6 discarded as burn-in ($6.7 \times 10^5$ samples, acceptance 0.38). The integrated autocorrelation time is $\tau \in [147, 183]$ steps, so the chain is $N/\tau \geq 113$ autocorrelation times long for every parameter, above the threshold of 50 [10] ($\gtrsim 3600$ effectively independent samples). Two runs started uniformly over the whole prior box converge to the same single mode (Appendix D.3, Fig. XIX). The profile likelihood is tabulated so that the result can be combined with others (Appendix D.4, Table VI).

### C. Posterior Results and Credible Intervals

The nominal model with the posterior medians coincides with the fit of Fig. II, which it follows over two orders of magnitude (Appendix D.5, Fig. XX). The marginal posterior on the raw asymmetry gives (median and equal-tailed credible intervals, CI)

$$\begin{aligned} A_{\mathrm{raw}} &= -0.0443, \\ 68\%\ \mathrm{CI}\ &[-0.0528, -0.0358], \\ 95\%\ \mathrm{CI}\ &[-0.0608, -0.0276]. \end{aligned}$$

The posterior is very close to Gaussian (standard deviation 0.0085), so equal-tailed and highest-posterior-density intervals coincide; with uniform priors its mode, $-0.0441$, is the maximum-likelihood estimate and the result can be compared directly with the Frequentist fit (Sec. IV). The other physics parameters (Fig. V) are $N_s = 15{,}328 \pm 133$, $N_b = 4404 \pm 82$, $A_b = -0.040 \pm 0.018$, $\mu = 5284.39 \pm 0.16$ MeV and $\sigma = 18.00 \pm 0.14$ MeV. $A$ is almost uncorrelated with them, which is why profiling (Frequentist) and marginalizing (Bayesian) the nuisance parameters give the same result; the largest correlation, $-0.17$, is with $A_b$: conditioning on $A_b = 0$ moves $A$ to $-0.0475$, as the fit with a charge-symmetric background does ($-0.0471$, Sec. IX).

### D. Bayes Factors

The Bayes factor compares two hypotheses through their evidences, $BF_{10} = P(D \mid H_1)/P(D \mid H_0)$, and requires proper priors. For the nested hypothesis $H_0$: $A = 0$ against $H_1$: $A$ free, with shared priors on the other parameters, it reduces to the Savage-Dickey ratio [11] (Appendix D.6),

$$BF_{01} = \frac{p(A = 0 \mid D)}{\pi(A = 0)}, \qquad \pi(A = 0) = \tfrac{1}{2}.$$

$A = 0$ lies 5.2 posterior standard deviations from the median, where the chain has essentially no samples, so $p(A = 0 \mid D)$ was taken from the Gaussian approximation of the marginal posterior: $\boldsymbol{BF_{01} \approx 1.1 \times 10^{-4}}$ ($BF_{10} \approx 9 \times 10^3$; Fig. VIa), **decisive evidence** against $A = 0$ on Jeffreys' scale [12]. The value scales with the prior width (Ockham factor): with $A \in (-0.3, 0.3)$, $BF_{01}$ drops by a factor 3.3 and the conclusion is unchanged (Appendix D.1, Fig. XVIII).

The same test on the background asymmetry shows the Ockham factor at work: $A_b = -0.040 \pm 0.018$ is 2.3 standard deviations from zero, yet $BF_{01}(A_b = 0) \approx 3.5$ (Fig. VIb), because the likelihood gain does not compensate for spreading the prior; unlike for $A$, the preference changes sign with the prior width. Of the Bayesian results *only the Bayes factor* depends on the prior; the interval of $A$ does not (Sec. V C). $A_b$ is kept free in the nominal model and fixing it to zero is one of the systematic variants (Sec. IX).

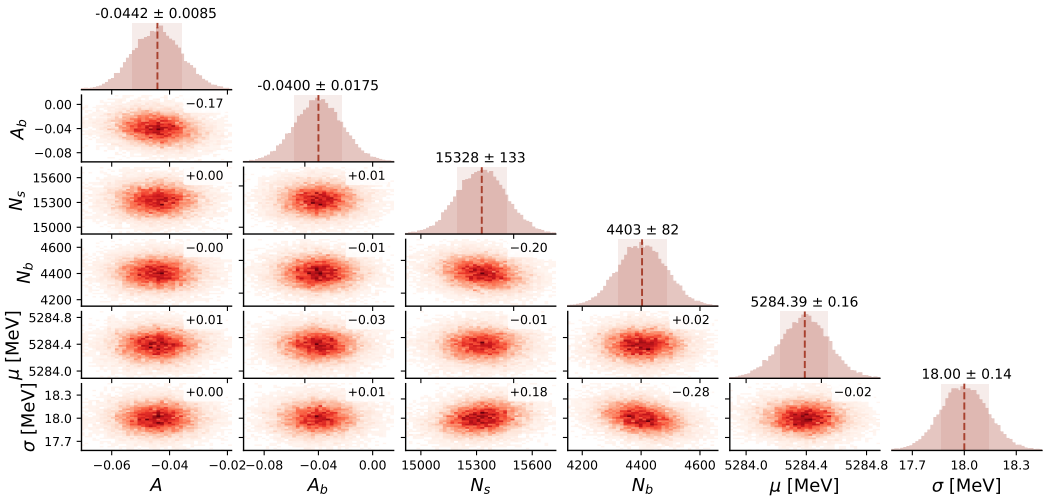



FIG. V. Marginal posteriors and correlations of the six physics parameters of the nominal fit (the six background-shape parameters are marginalized). Diagonal: posterior with its median (line) and 68% interval (band), quoted above as median ± half-width; below the diagonal: two-dimensional posteriors with the correlation coefficient.

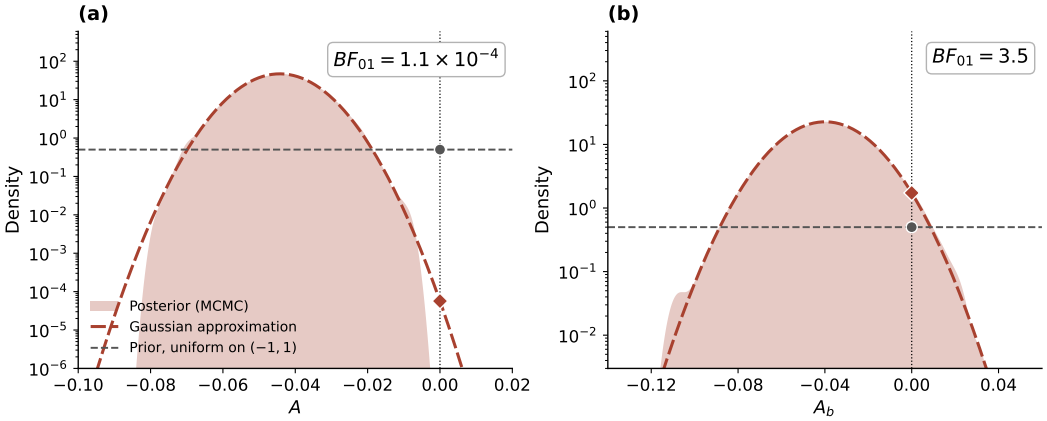



FIG. VI. Savage-Dickey ratio for (a) $A$ and (b) $A_b$: marginal posterior (histogram of the chain and Gaussian approximation) and prior density, logarithmic scale. The Bayes factor $BF_{01}$ is the ratio of the two densities at zero (dots). For $A$ the posterior density at zero is $10^{-4}$ times the prior one; for $A_b$, 2.3 standard deviations from zero, it is higher than the prior density, and the data favor $A_b = 0$.

### E. Additional Results

*Intervals.* Equal-tailed and highest-posterior-density 68% intervals coincide for $A$ and $A_b$ and differ for the skewed parameters; the profile likelihood is tabulated for combination with other results (Appendix D.7, Table VIII). *Priors.* No change of the prior on the nuisance parameters or of its parametrization moves $A$ by more than 0.0002, and the evidence by direct integration agrees in order of magnitude with the Savage-Dickey ratio (Appendix D.1, Table V). *Model adequacy.* The model does not reach $\chi^2/\mathrm{ndof} \approx 1$ (164.1/88), because the signal tails are heavier than a Gaussian, but $A_{\mathrm{raw}}$ changes by $-0.0002$ with a Crystal Ball signal; the posterior predictive check reproduces the asymmetry in slices of the mass ($p_B = 0.67$, the posterior predictive $p$-value, i.e. the fraction of replicas whose discrepancy exceeds the observed one) and puts the difference of the $\chi^2$ of the two charges at the edge of the replicas ($p_B = 0.08$) (Appendix D.5, Fig. XXI; Appendix D.8, Fig. XXII). *Robustness and coverage.* $A_{\mathrm{raw}}$ agrees between the magnet polarities ($0.7\sigma$) (Appendix D.9, Fig. XXIII); 200 toy-MC experiments give coverage $0.65 \pm 0.03$ (68%) and $0.93 \pm 0.02$ (95%) with pulls of width 1.04 (Appendix D.10, Fig. XXIV).

## VI. VALIDATION OF THE ESTIMATOR AND OF THE INTERVALS

*In short: toy-MC experiments show an unbiased estimator at the minimum variance bound, intervals with*

*nominal coverage and a test of $A = 0$ with the nominal size.* Toy-MC experiments generated from the fitted model of Sec. III (Poisson yields, masses drawn from the model by the transformation method; 1000 at the fitted value $A_{\text{true}} = -0.0441$ and 300 at each of $-0.1$, $-0.02$, 0 and $+0.05$, each fitted with the nominal procedure, 1–3% of the fits not valid and excluded) verify the estimator and the profile-likelihood ratio (Appendix E). Over the five values the bias of $\hat{A}$ is at most 0.0008 (1.6 standard errors; $-0.0005$, $-0.0001$, $-0.0008$, $+0.0002$, $-0.0004$ for the five values; Fig. VIIa), its spread equals the mean Hesse uncertainty within 6% and the pulls have width 0.94–1.05 (Fig. VIIb): the estimator is **unbiased** and reaches the **minimum variance bound**. At the fitted value the coverage of $t(A_{\text{true}}) \leq 1$ is $0.696 \pm 0.015$ (68.3% level) and $0.954 \pm 0.007$ (95.4% level); the 68% coverage over the five values is 0.71, 0.70, 0.69, 0.69 and 0.61. Inverting the Neyman construction, with 300 toy-MC experiments at each of 11 values of $A_{\text{true}}$ between $-0.1$ and 0, at the observed $\hat{A}$ (Appendix C.2, Fig. XV) gives the 68% interval $[-0.0528, -0.0351]$, against the Minos interval $[-0.0526, -0.0356]$. The test of $A = 0$ has the nominal size, $0.051 \pm 0.013$ at $A_{\text{true}} = 0$ against 0.05, and $q(0)$ is compatible with $\chi^2_1$ (Appendix E.1, Fig. XXVa); Figure VII shows (a) the bias, (b) the spread and the pull width and (c) the size and the power of the test as a function of the injected asymmetry.

### A. Additional Results

*Asymptotic behavior.* With the number of candidates the bias tends to zero, the spread scales as $N^{-1/2}$ and the distribution of $\hat{A}$ becomes Gaussian (Appendix E.2, Table IX). *Goodness of fit and binning.* The binned $\chi^2$ of toy experiments follows its asymptotic distribution, so its $p$-values are accurate: the misfit of the data is 5.4 standard deviations in 10 MeV bins and 2.0 in 2.5 MeV bins, because it lives on the scale of the flanks of the peak (Appendix E.3, Fig. XXVI).

## VII. MACHINE LEARNING

*In short: classifiers trained on weak labels from the mass spectrum reach a test AUC of* 0.86*; a cut on their score leaves $A_{\text{raw}}$ unchanged, and an adaptive partition of the Dalitz plane shows the pattern of the fixed binning.* The mass fits separate signal from background through the shape of the mass spectrum. As a complementary, data-driven approach, classifiers were trained to separate them from the other variables of the candidates, and used to test whether a machine-learning selection changes $A_{\text{raw}}$ and to partition the Dalitz plane without fixing the binning in advance.

### A. Weak Labels and Classifiers

No candidate carries a true label, so the classifiers were trained on weak labels derived from the mass spectrum (*weak labels*), following the classification-without-labels approach [13]: a classifier that separates two mixtures with different signal fractions, here the window $|m - \mu| < 2\sigma$ (15,130 signal-like candidates) and the sidebands $[5150, \mu - 4\sigma] \cup [\mu + 4\sigma, 5600]$ MeV (1940 background-like), is, by the Neyman-Pearson lemma, also the optimal signal/background classifier (it is monotonic in the signal-to-background ratio), provided that the features carry no information on the mass. The 20 features are the kinematic and particle-identification variables of the three tracks and two vertex variables of the $B$; the mass is not an input. Figure VIII shows the regions on the mass spectrum. The classes are imbalanced (0.128) and are balanced with class weights, and the sample is split 60/20/20 into training, validation and test sets. Six classifiers were compared: logistic regression with $L2$ and $L1$ penalty, a decision tree, a Random Forest, gradient boosting and a feed-forward neural network (FFNN). The tree ensembles and the network reach a test AUC of 0.857–0.864, 0.09 above the linear models and 0.04 above the single tree, and are indistinguishable from each other within the uncertainty of the test set (0.012); gradient boosting, the best in cross-validation, is the reference (Appendix F.1, Table X, Fig. XXVII).

### B. Regularization, Bias-Variance and Uncertainty

The regularization strength of the logistic regression (Appendix F.2), scanned over 30 values of $C = 1/\alpha$ with cross-validation, gives the same optimal AUC for the $L1$ and $L2$ penalties (0.7686) and no overfitting: the limitation of the linear model is its decision boundary, not the variance (Appendix F.3, Fig. XXIX). For gradient boosting, whose complexity grows with the number of stages, the test AUC rises up to $\sim$ 150 stages and then stays at 0.865 while the training AUC keeps rising (the gap grows from 0.018 at 20 stages to 0.084 at 300): the additional stages overfit without gain (Appendix F.3, Fig. XXX). The uncertainty of a prediction was estimated from the spread of the 300 trees of the Random Forest, of 300 forward passes with dropout kept active (MC dropout) and of a deep ensemble of 10 networks: the mean per-candidate values are 0.178, 0.095 and 0.050; they measure different sources of variability, and only the Random Forest spread decreases visibly with the size of the training set (0.220, 0.204, 0.179 with a quarter, half and all of it), the epistemic part, while the part at scores around 0.5 is the aleatoric one (Appendix F.4, Fig. XXXII).

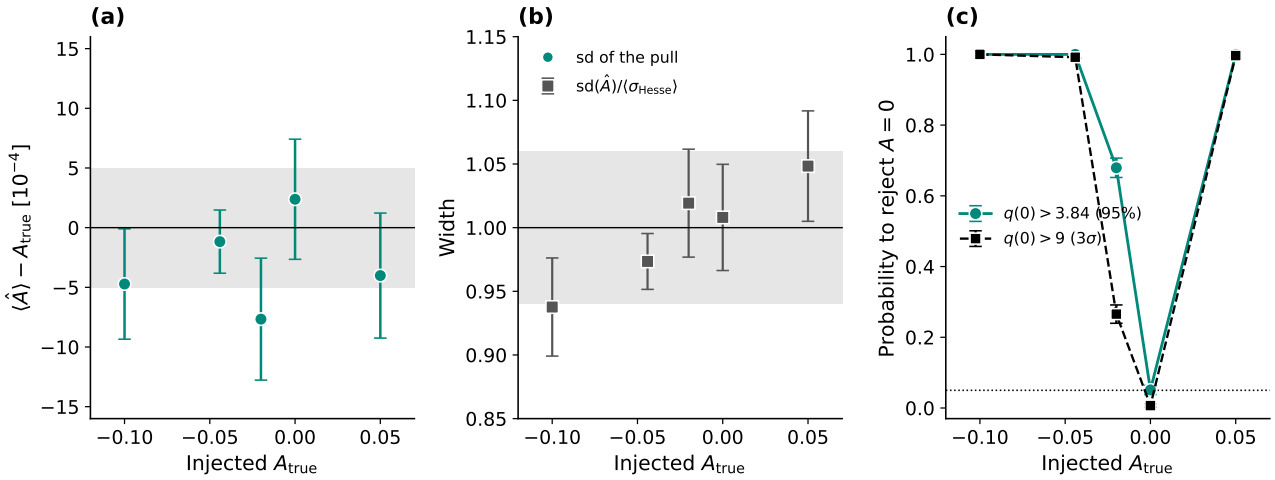



FIG. VII. Toy-MC experiments at five injected asymmetries. (a) Bias of $\hat{A}$, with its standard error. (b) Spread of the estimates divided by the mean Hesse uncertainty, and width of the pulls. (c) Probability to reject $A = 0$ in the test with $q(0) > 3.84$ (size 5%) and $q(0) > 9$ ($3\sigma$), as a function of the injected value: size at $A_{\rm true} = 0$, power elsewhere.

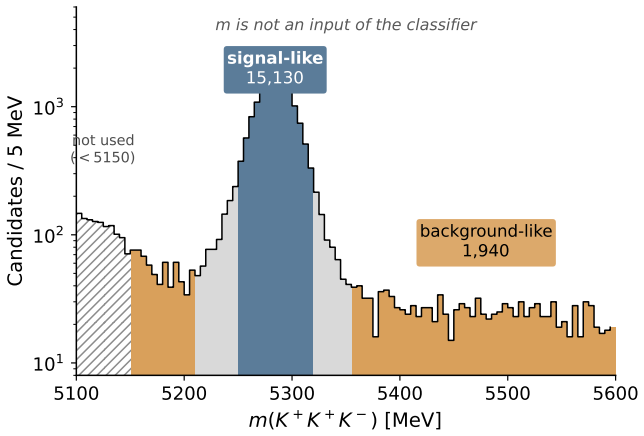



FIG. VIII. Weak labels on the mass spectrum of the fit window: signal-like candidates ($|m - \mu| < 2\sigma$, 15,130) and background-like candidates (sidebands beyond $4\sigma$ and above 5150 MeV, 1940); the 2–$4\sigma$ gap is left unlabeled and the region below 5150 MeV, where the background rises towards partially reconstructed decays, is excluded.

### C. Effect on the Measurement and Dalitz Partition

A cut at 0.5 on the score keeps 92% of the fitted signal and 36% of the background (15,709 candidates). The raw asymmetry is **stable** (Fig. X): $-0.0451 \pm 0.0087$ (Bayesian) and $-0.0447 \pm 0.0086$ (Frequentist), against $-0.0443$ and $-0.0441$ for the full window, all shifts within the spread expected for a subsample and $5.2\sigma$ from zero in both samples. The precision does not improve, because the nominal sample has little background to remove ($\approx 4\%$ in the signal window). The mass spectrum, instead, changes ($\chi^2$ from 164/88 to 253/88 in the Bayesian fit): the classifier does not sculpt the mass, since without the nine momentum components the degradation is the same ($\chi^2 = 533/388$ in the Frequentist fit), but the selection changes the composition of the background, keeping partially reconstructed decays that the model does not describe (Appendix F.5, Table XI, Fig. XXXIV). Finally, $K$-means with $K = 8$ on $(m^2_{\rm lo}, m^2_{\rm hi})$, an *unsupervised* partition learned from the density of the candidates and stable between initializations (Appendix F.6), finds two clusters beyond $3\sigma$ ($\chi^2 = 46.4$ for 7 degrees of freedom) with the pattern of the fixed binning ($\chi^2 = 80.8$ for 31 degrees of freedom, Sec. VIII): negative at low $m^2_{\rm lo}$ ($-0.127 \pm 0.017$) and positive at high $m^2_{\rm hi}$ (Fig. IX) (Appendix F.7, Fig. XXXVI).

## VIII. LOCAL ASYMMETRY IN THE DALITZ PLANE

*In short: the asymmetry is not uniform over the Dalitz plane ($\chi^2 = 80.8$ for 31 degrees of freedom) and is concentrated outside the $\phi(1020)$, in the region of the published analysis: $-0.234 \pm 0.025$ against $-0.239 \pm 0.020$.* To test whether the asymmetry is uniform over phase space or concentrated in specific regions, each candidate was placed in the Dalitz plane using the two opposite-sign $K^+K^-$ invariant masses $m^2(H1, H2)$ and $m^2(H1, H3)$ (Sec. II A), sorted into $m^2_{\rm lo}$ and $m^2_{\rm hi}$ to remove the $H2/H3$ labeling ambiguity. The plane $[0, 16] \times [0, 30]$ GeV$^2$ in $(m^2_{\rm lo}, m^2_{\rm hi})$ was divided into an $8 \times 8$ grid, fixed before looking at the asymmetries. In each bin with at least 20 candidates (32 of 64), the local asymmetry $A_i = (N_i^- - N_i^+)/N_i$, with binomial uncertainty $\sigma(A_i) = \sqrt{(1 - A_i^2)/N_i}$, was compared with the global counting asymmetry $A_{\rm glob} = -0.043$ through $S_i = (A_i - A_{\rm glob})/\sigma(A_i)$.

Figure XI shows the occupancy of the bins (a), the lo-

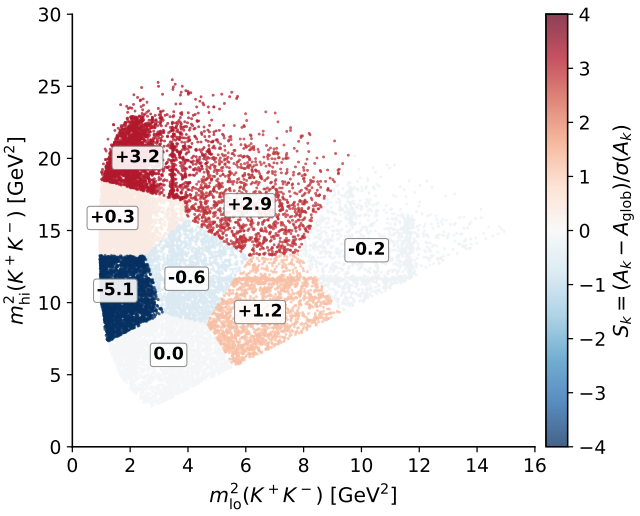



FIG. IX. $K$-means partition ($K = 8$) of the Dalitz plane, all candidates of the fit window, same axes as Fig. XI. Color and labels: local significance of each cluster with respect to the global asymmetry, $S_k = (A_k - A_{\rm glob})/\sigma(A_k)$, on the same scale as panel (c) of Fig. XI.

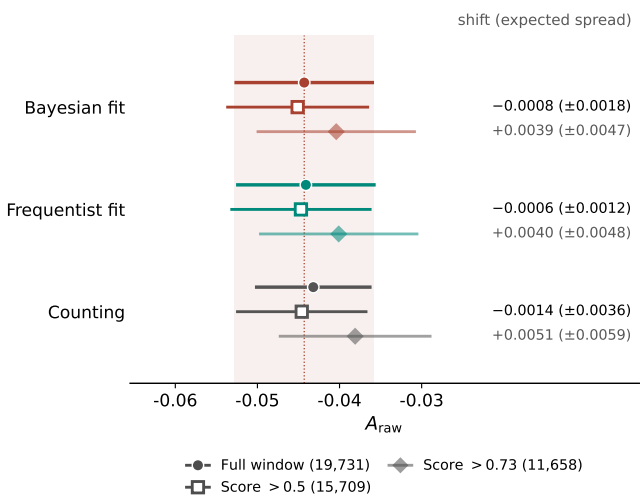



FIG. X. $A_{\rm raw}$ from the Bayesian fit, the Frequentist fit and counting, for the full sample and the ML-selected ones, at a score cut of 0.5 and at 0.73, the maximum of the significance improvement.

cal asymmetry (b) and its significance (c). Two of the 32 bins have $|S_i| > 3$, adjacent at $m^2_{\rm lo} < 4$ GeV$^2$ and $7.5 < m^2_{\rm hi} < 11.25$ GeV$^2$ ($S_i = -3.7$ and $-3.0$); about 0.09 such bins would be expected if the asymmetry were uniform, and two or more with probability $3 \times 10^{-3}$, a look-elsewhere number over any two of the 32 bins. The global homogeneity test, $\chi^2 = \sum_i S_i^2 = 80.8$ for 31 degrees of freedom ($p = 3 \times 10^{-6}$), confirms that the local asymmetry is **not uniform**; it also holds in the signal window alone, with the charm vetoes and for other grids (Appendix G.1).

The structure is the same with both magnet polarities (Appendix G.1): the split tests the part of the detection asymmetry that changes sign with the polarity, not the different interaction of $K^+$ and $K^-$ with the detector material. Where is the asymmetry in terms of the decay dynamics? The Dalitz plane shows the $\phi(1020)$ band at the kinematic edge ($m^2_{\rm lo} \approx 1.04$ GeV$^2$) and the charm bands of the $D^0$, $J/\psi$ and $\chi_{c0}$. The negative asymmetry is *not* produced by the $\phi(1020)$: its band (1964 candidates) has $A_{\rm count} = -0.004 \pm 0.023$, while the significant bins without it have $-0.145 \pm 0.019$ (Appendix G.2, Fig. XXXVIII). It sits just above the $\phi$, where the published analysis located it, defining the region $1.2 < m^2_{\rm lo} < 2.0$ GeV$^2$, $m^2_{\rm hi} < 15$ GeV$^2$ with $A_{\rm raw} = -0.239 \pm 0.020$ [5]. The region is fixed by the paper, not chosen on our data, so the comparison is unbiased: the nominal model fitted to its 2146 candidates gives $A_{\rm raw} = \mathbf{-0.234 \pm 0.025}$ (Bayesian) and $-0.237 \pm 0.025$ (Frequentist, $9.4\sigma$ from zero), reproducing the published value.

### A. Additional Results

*Null tests.* Permuting the charges gives $\chi^2 = 31.5 \pm 8.4$ and never reaches 80.8 in 2000 permutations, and the phase-space simulation is consistent with no asymmetry (slightly optimistic $p$-values) (Appendix G.3, Fig. XL).

## IX. RESULT AND SYSTEMATIC UNCERTAINTIES

*In short: thirteen variants in four groups, fitted by both methods on the same candidates, give* 0.0070 *(Bayesian) and* 0.0072 *(Frequentist), dominated by the particle-identification requirement.* The result depends on choices of the analysis that the data do not fix. They were varied one at a time in thirteen variants, fitted by both methods on exactly the same candidates and combined with the same rule. The variants fall in four groups: the *window* (lower edge 5050 and 5150 MeV, upper edge 5450 and 5700 MeV), the *background* model (single exponential, shape shared between the charges, charge-symmetric background $A_b = 0$), the *signal* model (Crystal Ball, $\mu$ and $\sigma$ per charge) and the *preselection* (`ProbK` $> 0.4$ and 0.6, `ProbPi` $< 0.4$ and 0.6). When a variant changes the sample (window, preselection) the two samples are nested and part of the shift $\Delta = A^{\rm var}_{\rm raw} - A^{\rm nom}_{\rm raw}$ is a statistical fluctuation of expected size $\sigma_B = \sqrt{|\sigma^2_{\rm var} - \sigma^2_{\rm nom}|}$ [14] (the difference of two nearly equal uncertainties, hence sensitive to small differences between the methods: a variant at the threshold can be classified differently without changing the result), so the variant contributes only if $|\Delta| > \sigma_B$; a variant that the data reject (the single exponential) is excluded; the largest remaining $|\Delta|$ of each group is its uncertainty and the groups, being independent choices, are added in quadrature (Appendix H.1,

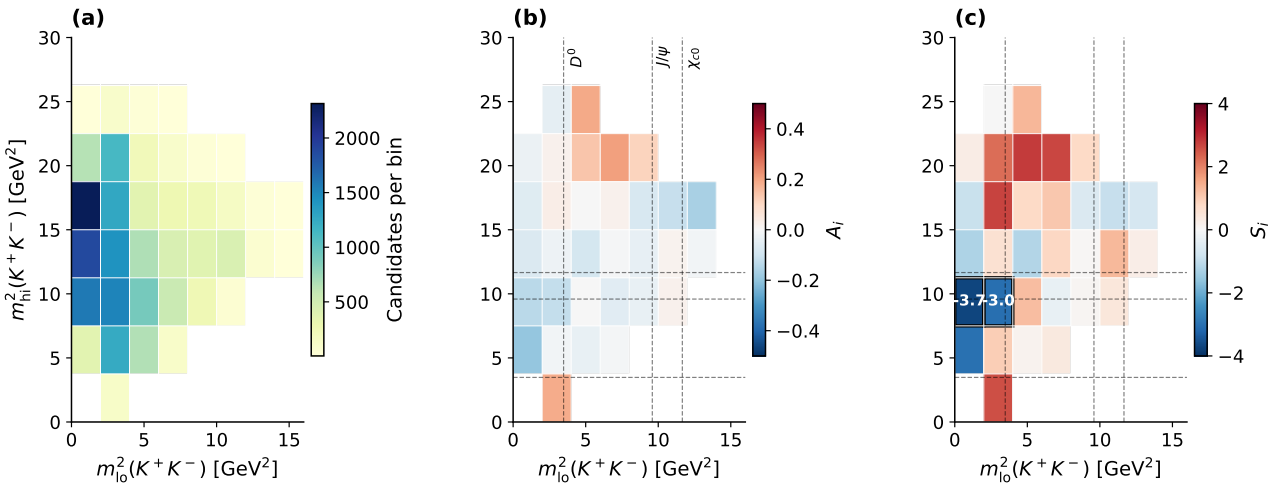



FIG. XI. Fixed $8\times8$ Dalitz-plot binning, all candidates in the fit window: (a) number of candidates per bin, (b) local asymmetry $A_i$ for bins with $\geq 20$ candidates, and (c) local significance with respect to the global asymmetry, $(A_i - A_{\rm glob})/\sigma(A_i)$. The two bins with $|S_i| > 3$ are outlined. Dashed lines: the squared masses of the $D^0$, $J/\psi$ and $\chi_{c0}$, in either opposite-sign pair.

Tables XIV and XV). The groups give 0.0014 (window), 0.0037 (background), 0.0002 (signal) and 0.0058 (preselection) in the Bayesian fit and 0.0014, 0.0040, 0.0006 and 0.0058 in the Frequentist one, for totals of 0.0070 and 0.0072; the two methods give the same shifts to within $\sim$0.0005. The **preselection** dominates, through the single variant `ProbK` $> 0.6$ ($+0.0058$, $1.2\,\sigma_B$), which a scan of each cut in steps of 0.05 shows to be a *fluctuation*: on the tight side the shifts scatter around zero within $1.3\,\sigma_B$ without a trend, whereas loosening `ProbK` moves $A_{\rm raw}$ monotonically, as expected if misidentified decays not in the mass model enter the sample (Fig. XII). The nominal requirement is therefore well placed and the preselection uncertainty conservative. The shifts of the thirteen variants are shown in Fig. XIII, and the final result is

$$\text{Bayesian:}$$
$$A_{\rm raw} = -0.0443 \pm 0.0085\,(\text{stat}) \pm 0.0070\,(\text{syst}),$$
$$\text{Frequentist:}$$
$$A_{\rm raw} = -0.0441 \pm 0.0085\,(\text{stat}) \pm 0.0072\,(\text{syst}).$$

### A. Additional Results

*Checks.* The $\sigma_B$ rule is verified by bootstrap (the spread of the shift between a sample and its tighter subsample is $\sigma_B$), and the grid scans of the Frequentist fit move $A_{\rm raw}$ by at most 0.0126 (PID cuts, smaller samples), 0.0048 (window edges) and $6\times10^{-5}$ (signal region) (Appendix H.2, Fig. XLI; Appendix H.3, Fig. XLII).

## X. DISCUSSION AND CONCLUSIONS

*In short: the Frequentist and Bayesian analyses give the same* $A_{\rm raw} = -0.044 \pm 0.009\,(stat) \pm 0.007\,(syst)$, *5.2 standard deviations from zero, with validated intervals; they differ in the meaning of the numbers and of the tests, not in their values.* On this problem the Frequentist profile-likelihood fit and the Bayesian posterior are equivalent in value, uncertainty and coverage: with a common selection and mass model they give $A_{\rm raw} = -0.0441 \pm 0.0085 \pm 0.0072$ and $-0.0443 \pm 0.0085 \pm 0.0070$ (statistical, systematic), a nonzero asymmetry at 5.2 standard deviations, stable under the thirteen variants (the dominant systematic contribution is the particle-identification requirement) and concentrated in the Dalitz region of the published analysis outside the $\phi(1020)$ ($-0.234 \pm 0.025$ against $-0.239 \pm 0.020$). **The numbers coincide, the statements do not.** The *credible interval* states that $A$ lies in $[-0.0528, -0.0358]$ with 68% probability given the data and the prior; the *confidence interval* $[-0.0526, -0.0356]$ is the set of values that the data do not reject, and its 68% is a property of the procedure that the bootstrap and the toy experiments verified. The Frequentist test of $A = 0$, $p = 2\times10^{-7}$, is independent of the prior; the Bayes factor, $BF_{10} \approx 9 \times 10^3$, depends on its width (Sec. V D), and for $A_b = -0.040 \pm 0.018$ a Frequentist test would call $2.3\sigma$ a hint while $BF_{01} \approx 3.5$ mildly favors $A_b = 0$. The Frequentist fit adds an unbiased estimator at the minimum variance bound; the Bayesian analysis adds sampling diagnostics and a posterior predictive check. Figure XIV shows the profile likelihood and the marginal posterior of $A$ on the same axis, and the likelihoods of the two magnet polarities.

The published analysis [5] finds $-0.060 \pm 0.007$ on a larger sample ($22{,}119 \pm 164$ signal candidates) with a richer model (a Cruijff signal and misidentified and par-

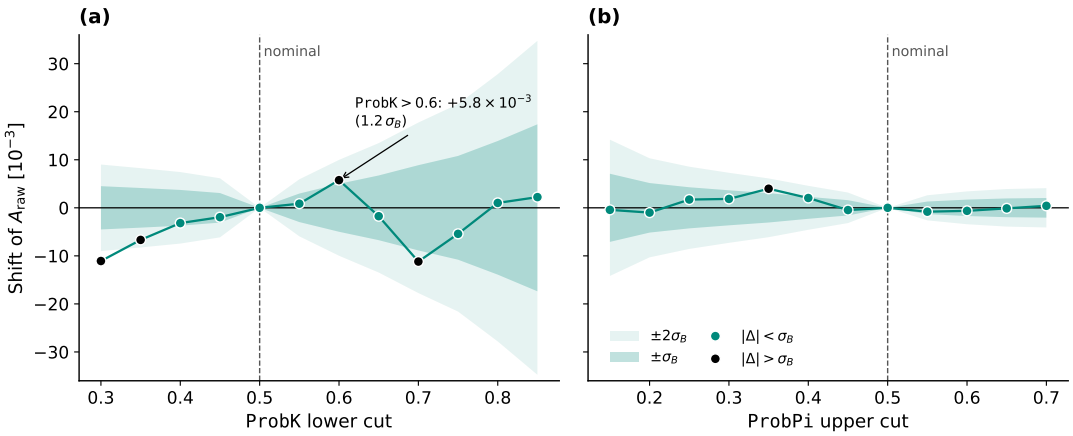



FIG. XII. Shift of $A_{\rm raw}$ (Frequentist fit) as a function of (a) the `ProbK` lower cut and (b) the `ProbPi` upper cut, in steps of 0.05, with the expected statistical spread of the nested sample ($\pm\sigma_B$ dark, $\pm2\sigma_B$ light; zero at the nominal cut, dash-dotted line). On the tight side of `ProbK` the points scatter around zero without trend and the pion requirement is flat; loosening `ProbK` moves $A_{\rm raw}$ monotonically.

tially reconstructed backgrounds); with the same $D^0$ veto the Bayesian fit gives $-0.0496 \pm 0.0088$, within $0.93\sigma$. The raw asymmetry is not the $CP$ asymmetry, $A_{CP} \approx A_{\rm raw} - A_P - A_D$ (Eq. (2); in the published analysis the corrections move $-0.060$ to $-0.043 \pm 0.009 \pm 0.003 \pm 0.007$ and cannot be assumed for a different selection): the $5.2\sigma$ here is the significance of a nonzero raw asymmetry, not of $CP$ violation. Four limitations bound the result: no tested model reaches $\chi^2/\text{ndof} \approx 1$, although $A_{\rm raw}$ does not depend on the signal shape; no production or detection correction is applied (the record has no control channel, and its other three channels are not analysed); the local Dalitz-plane asymmetries are counts without background subtraction; and the machine-learning labels are weak and imbalanced. A machine-learning selection leaves $A_{\rm raw}$ unchanged within its statistical spread (Sec. VII), and the natural next step is a control-channel correction.

### A. Additional Results

*Numerical agreement.* The estimates of the two fits differ by 0.0002, both uncertainties are 0.0085, and sideband subtraction gives $-0.0503 \pm 0.0083$. *Profile, posterior and polarities.* Profiling and marginalizing the eleven nuisance parameters give the same distribution of $A$, since $A$ is almost uncorrelated with them; the two polarities agree and their combined likelihood reproduces the joint fit; the 68% and 95% regions in $(A, A_b)$ of the profile ($\Delta$NLL at 2.30 and 5.99) and of the highest posterior density coincide, with $A = 0$ outside both (Appendix I.1, Table XVI; Appendix I.2, Fig. XLIII). *Tests and goodness of fit.* The exact binomial test of $N^- = N^+$ on all 19,731 candidates ($A_{\rm count} = -0.0432 \pm 0.0071$) gives $6.1\sigma$, larger than the $5.2\sigma$ of the fit ($-0.0441 \pm 0.0085$), and the two are not in tension: the count tests the charge asymmetry of signal and background together, without the mass, so its uncertainty does not include the cost of separating the background, while the fit tests the asymmetry of the signal alone with that of the background free; the $5.2\sigma$ is the one that refers to the signal. The same fit has $p = 0.03$ in 2.5 MeV bins and $p \approx 5 \times 10^{-6}$ in 10 MeV bins: the $p$-value of the binned $\chi^2$ depends on the binning as much as on the model (Appendix E.3, Fig. XXVI). *Low statistics.* Credible and confidence intervals coincide in value and coverage from $\sim 2 \times 10^4$ down to 50 candidates (Appendix I.3, Fig. XLIV). *Machine learning: hyperparameters and features.* The hyperparameter search gains at most 0.005 in AUC, less than its uncertainty, and the cross-validated maximum overestimates the test AUC (winner's curse); the Random Forest importance and the Lasso coefficients of the features agree only partly and both rank the two vertex variables of the $B$ among the top five (Appendix F.8, Fig. XXXVII, Table XII; Appendix F.1, Fig. XXVIII).

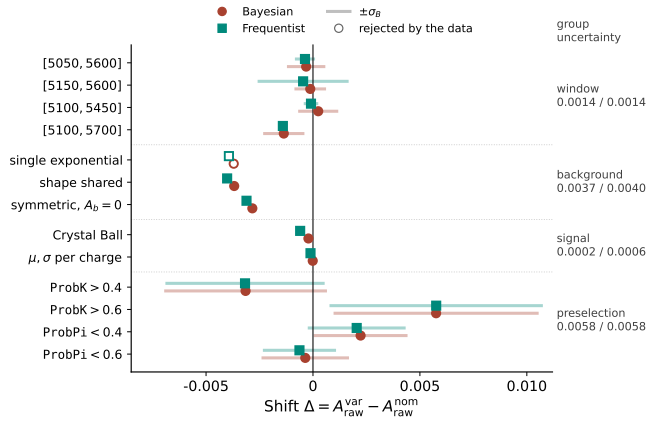



FIG. XIII. Shift $\Delta$ of $A_{\text{raw}}$ for the thirteen variants of the analysis, in the Bayesian (circles) and Frequentist (squares) fits, with the expected statistical spread $\sigma_B$ of the variants that change the sample (bars); the open markers are the variant rejected by the data. On the right, the uncertainty of each group (Bayesian / Frequentist), its largest contributing shift.

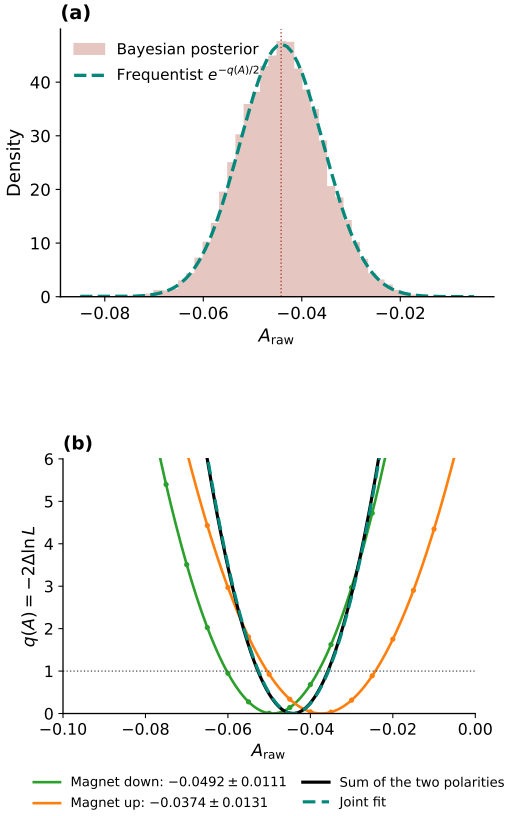



FIG. XIV. (a) Frequentist profile likelihood, $\exp[-q(A)/2]$ normalized, and Bayesian marginal posterior of $A$: the two coincide. (b) Profile likelihood $q(A)$ of the two magnet polarities (parabolic fits to the points), their sum and the joint fit: the likelihoods of the two polarities combine into that of the full sample.

## Appendix A: Dataset and Preselection

*In short: the branches read by the analysis and the cut flow of the preselection.*

**1. Branches and Preselection** *In short: the analysis reads momentum, particle-identification likelihoods, muon flag and charge of each track; the common preselection keeps* 23,895 *candidates (*0.28%*), mostly through the kaon-identification requirement.*

**2. Cut Flow of the Preselection** *In short: the kaon-identification requirement on the three tracks is by far the most selective (Table II).*

Table II gives the number of candidates passing each requirement alone and cumulatively, by magnet polarity.

| Requirement | alone | cumulative | down | up |
|---|---|---|---|---|
| All candidates | – | 8,556,118 | 5,135,823 | 3,420,295 |
| ¬`isMuon`, three tracks | 6,311,517 | 6,311,517 | 3,806,297 | 2,505,220 |
| `ProbPi` $< 0.5$, three tracks | 975,724 | 481,101 | 288,908 | 192,193 |
| `ProbK` $> 0.5$, three tracks | 28,812 | 23,895 | 14,178 | 9717 |
| Mass window $[5100, 5600]$ MeV | – | 19,731 | 11,636 | 8095 |

TABLE II. Number of candidates passing each requirement alone and after the cumulative selection, in the order of the table, and by magnet polarity. The kaon-identification requirement on the three tracks is by far the most selective.

### Appendix B: Mass Model and Likelihood

*In short: the relation between the extended likelihood and a mixture with fixed total.*

**1. Extended Likelihood versus a Mixture with Fixed Total** *In short: the extended form differs from a mixture with fixed total only by the Poisson term of the total yield, which carries no information on A; the purity of the signal window is about* 96%.

### Appendix C: Frequentist Analysis

*In short: why the Hesse matrix and the significance $Z = \sqrt{q(0)}$ follow from the likelihood, the Neyman construction, and the uncertainties of the twelve parameters by three methods.*

**1. Hesse Uncertainties, Test of $A = 0$ and Significance** *In short: the Hesse matrix is the inverse of the Fisher information, the test statistic is the profile likelihood ratio, and the significance is $Z = \sqrt{q(0)}$. Why $Z = \sqrt{q(0)}$.* Let $\theta$ be all the parameters, $\hat{\theta}$ the maximum-likelihood estimate and $I$ the Fisher information. Near the maximum the log-likelihood is parabolic,

$$-2\ln L(\theta) \simeq -2\ln L(\hat{\theta}) + (\theta-\hat{\theta})^T I\,(\theta-\hat{\theta}),$$

so asymptotically $\hat{\theta}$ is Gaussian with covariance $V = I^{-1}$: this is the Hesse matrix, and $\sigma(A) = \sqrt{V_{AA}}$ already includes the effect of the other parameters. For the null hypothesis $A = 0$ the statistic is the profile likelihood ratio $q(0) = -2\ln\big[L(0,\hat{\hat{\theta}})/L(\hat{A},\hat{\theta})\big]$. For a parabolic likelihood, maximizing over the other parameters at fixed $A$ gives

$$q(0) = \frac{\hat{A}^2}{V_{AA}} = \Big(\frac{\hat{A}}{\sigma(A)}\Big)^2,$$

and under $H_0$ the variable $\hat{A}/\sigma(A)$ is a standard normal $N$. Hence $q(0) = N^2 \sim \chi^2_1$: one degree of freedom, the parameter fixed by $H_0$, whatever the number of the others. This is Wilks' theorem, exact for a parabolic likelihood and an approximation otherwise. The $p$-value is then

$$p = P(\chi^2_1 \geq q_{\rm obs}) = P\big(|N| \geq \sqrt{q_{\rm obs}}\big) = 2\big[1-\Phi(\sqrt{q_{\rm obs}})\big],$$

which is the probability of a normal variable beyond $\sqrt{q}$ standard deviations: $Z = \sqrt{q(0)}$ is the significance in units of $\sigma$ (two-sided). With $q \simeq 27$, $p = 2.05\times10^{-7}$ and $Z = 5.20$; the one-sided convention $\Phi^{-1}(1-p)$ would give 5.06, and the Hesse ratio $|\hat{A}|/\sigma(A) = 5.19$ coincides with $\sqrt{q}$ because the likelihood is nearly parabolic. The approximation fails if the sample is small or $A$ is near a boundary of its range.

**2. Neyman Construction and Dependence on the Momentum** *In short: inverting the test of $A_{\rm true}$ with toy-MC experiments gives the same interval as the likelihood ratio (Fig. XV), and $A_{\rm raw}$ does not depend on the momentum of the B (Table III, Fig. XVI).*

Figure XV shows the central 68% and 95% ranges of $\hat{A}$ for each $A_{\rm true}$: the interval is read where the observed $\hat{A}$ crosses the 68% band.

Table III gives $A_{\rm raw}$ in terciles of the transverse momentum and of the momentum of the $B$ candidate.

Figure XVI shows the same values in terciles of (a) the transverse momentum and (b) the momentum, against the global value and its $\pm1\sigma$ band.

**3. Uncertainties of the Twelve Parameters by Three Methods** *In short: the inverse of the Fisher information (Hesse), the bootstrap of the data and the posterior give the same standard deviations for A, $A_b$, $\mu$ and $\sigma$ (Table IV, Fig. XVII); the bootstrap differs for the two yields, because it keeps the total number of candidates fixed, and for the steep slope $\lambda_1^+$, which is weakly constrained.*

Table IV lists the standard deviation of each parameter of the nominal fit by the three methods.

Figure XVII shows the standard deviations from the bootstrap (300 replicas) and from the posterior divided by the Hesse uncertainty, with a $\pm10\%$ band.

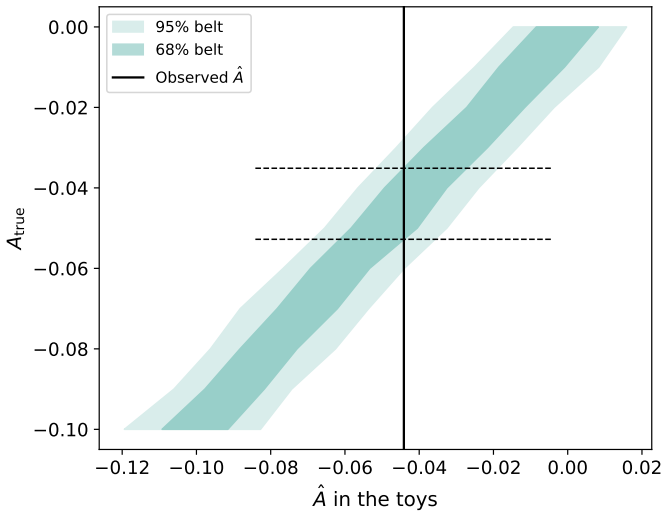



FIG. XV. Confidence belt of $\hat{A}$ from toy-MC experiments (central 68% and 95% ranges) as a function of $A_{\text{true}}$; the vertical line is the observed $\hat{A}$ and the dashed lines the resulting 68% interval.

| Bin | range [GeV] | $n$ | $A_{\text{raw}}$ |
|---|---|---|---|
| $p_T$, 1 | 1.0–4.7 | 6577 | $-0.0563 \pm 0.0169$ |
| $p_T$, 2 | 4.7–8.2 | 6577 | $-0.0362 \pm 0.0141$ |
| $p_T$, 3 | 8.2–30.5 | 6577 | $-0.0452 \pm 0.0134$ |
| $p$, 1 | 26.7–78.1 | 6577 | $-0.0391 \pm 0.0139$ |
| $p$, 2 | 78.1–110.1 | 6577 | $-0.0475 \pm 0.0149$ |
| $p$, 3 | 110.1–288.5 | 6577 | $-0.0478 \pm 0.0153$ |

TABLE III. $A_{\text{raw}}$ (joint frequentist fit) in terciles of the transverse momentum and of the momentum of the $B$ candidate.

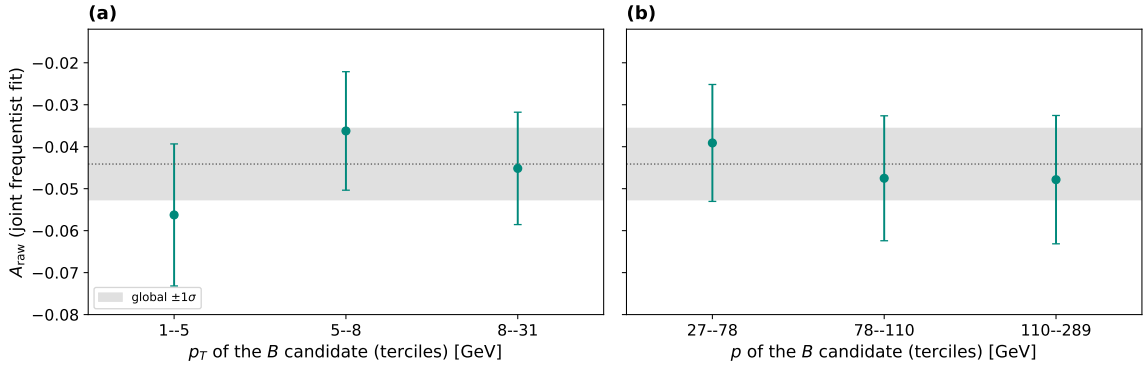



FIG. XVI. $A_{\text{raw}}$ (joint frequentist fit) in terciles of (a) the transverse momentum and (b) the momentum of the $B$ candidate, with the global value and its $\pm 1\sigma$ band; each bin has 6577 candidates.

| Parameter | Hesse | Bootstrap | Posterior |
|---|---|---|---|
| $A$ | 0.0085 | 0.0085 | 0.0085 |
| $A_b$ | 0.0177 | 0.0183 | 0.0176 |
| $N_s$ | 133 | 72 | 133 |
| $N_b$ | 82 | 72 | 82 |
| $\mu$ [MeV] | 0.159 | 0.159 | 0.161 |
| $\sigma$ [MeV] | 0.141 | 0.150 | 0.138 |
| $\lambda_1^+$ | 0.0077 | 0.0057 | 0.0078 |
| $\lambda_2^+$ | 0.00028 | 0.00030 | 0.00029 |
| $f_+$ | 0.028 | 0.029 | 0.030 |
| $\lambda_1^-$ | 0.0047 | 0.0043 | 0.0047 |
| $\lambda_2^-$ | 0.00040 | 0.00038 | 0.00039 |
| $f_-$ | 0.039 | 0.036 | 0.037 |

TABLE IV. Standard deviation of the parameters of the nominal fit from the Hesse matrix, from the bootstrap of the data and from the posterior.

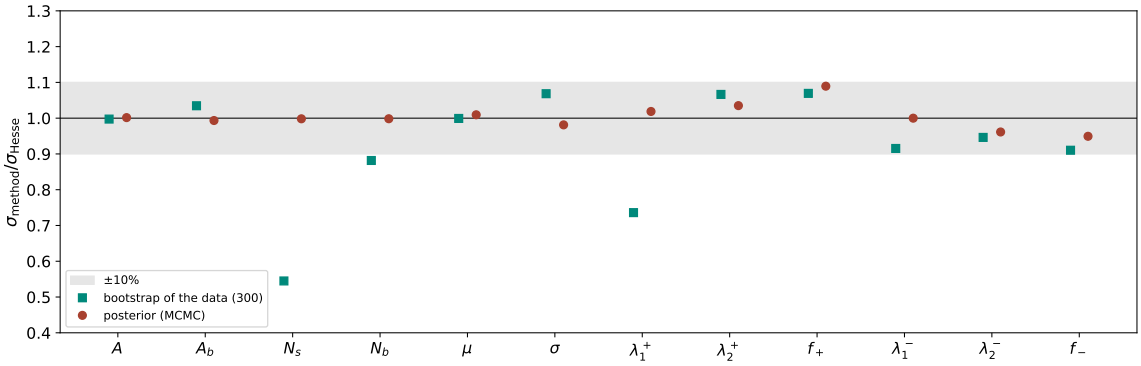



FIG. XVII. Standard deviation of the twelve parameters of the nominal fit from the bootstrap of the data (300 replicas) and from the posterior, divided by the Hesse uncertainty; the band is ±10%.

## Appendix D: Bayesian Analysis

*In short: the prior sensitivity, the sampler and its trace plots, the derivation of the Savage-Dickey ratio, the intervals, the model-adequacy checks, the magnet-polarity check, the toy-MC coverage of the credible interval and the tabulated profile likelihood.*

**1. Priors and Bayes Factors** *In short: no change of the prior on the nuisance parameters or of its parametrization moves $A$ by more than 0.0002 (Table V); the Bayes factor of $A = 0$ scales with the prior width (Fig. XVIII), and the evidence computed by direct integration agrees in order of magnitude with the Savage-Dickey ratio.*

Table V gives the posterior of $A$ under the alternative priors, with the effective sample size of the reweighted chain.

| Prior | $A$ (median) | 68% interval | ESS |
|---|---|---|---|
| Nominal (uniform) | $-0.0442$ | $[-0.0529, -0.0358]$ | 20000 |
| Log-uniform $N_s$, $N_b$ | $-0.0442$ | $[-0.0529, -0.0358]$ | 19993 |
| Log-uniform slopes | $-0.0440$ | $[-0.0528, -0.0356]$ | 12306 |
| Uniform in $1/\lambda$ | $-0.0444$ | $[-0.0526, -0.0360]$ | 153 |
| Flat in $(N_s^+, N_s^-)$ | $-0.0442$ | $[-0.0529, -0.0358]$ | 19999 |
| Flat in $(N_s^\pm)$ and $(N_b^\pm)$ | $-0.0442$ | $[-0.0529, -0.0358]$ | 19993 |

TABLE V. Posterior of $A$ under alternative priors, by importance reweighting of the chain; ESS: effective sample size of the 20,000 thinned samples.

Figure XVIII shows $BF_{01}$ of $A = 0$ and of $A_b = 0$ against the half-width of the uniform prior, with the nominal prior marked.

**2. Sampling the Posterior** *In short: Metropolis-Hastings accepts a proposal with a ratio that satisfies detailed balance; the affine-invariant ensemble move samples correlated posteriors without tuning, and $N/\tau$ is the effective length of a chain.*

**3. Trace Plots of the Walkers** *In short: walkers started over the whole prior box converge to the same mode (Fig. XIX).*

Figure XIX shows the trace of $A$, $N_s$, $\sigma$ and $f_+$ for the 32 walkers: the first 3000 steps over the full range, and the whole chain zoomed on the posterior with the burn-in marked.

**4. Published Profile Likelihood** *In short: the profile likelihood ratio $q(A)$ of the nominal fit is tabulated (Table VI), so that the result can be combined with others.*

Table VI gives $q(A)$ for $A$ between $-0.08$ and 0, with the minimum and the Minos intervals of the nominal fit (other parameters profiled).

**5. Model Adequacy: Goodness of Fit and Posterior Predictive Check** *In short: the model does not reach $\chi^2/\mathrm{ndof} \approx 1$ (164.1/88), mainly because the signal tails are heavier than a Gaussian, but $A_{\mathrm{raw}}$ changes by only $-0.0002$ with a Crystal Ball signal (Table VII); the pulls of the mass fit are in Fig. XX.*

| $A$ | $-0.08$ | $-0.07$ | $-0.06$ | $-0.05$ | $-0.04$ | $-0.03$ | $-0.02$ | $-0.01$ | $+0.00$ |
|---|---|---|---|---|---|---|---|---|---|
| $q(A)$ | 17.9 | 9.3 | 3.5 | 0.5 | 0.2 | 2.8 | 8.1 | 16.1 | 27.0 |

TABLE VI. Profile likelihood ratio $q(A) = -2\Delta \ln L$ of the nominal fit (other parameters profiled); minimum at $-0.0441$, Minos intervals $[-0.0526, -0.0356]$ (68%) and $[-0.0607, -0.0275]$ (95%).

| Model | $A_{\mathrm{raw}}$ | $\chi^2/\mathrm{ndof}$ | $p$ |
|---|---|---|---|
| Nominal | $-0.0443$ | 164.1/88 | $2 \times 10^{-6}$ |
| Crystal Ball signal | $-0.0445$ | 131.0/86 | $1 \times 10^{-3}$ |
| $\mu$, $\sigma$ per charge | $-0.0443$ | 164.2/86 | $8 \times 10^{-7}$ |
| Shared bkg. shape | $-0.0480$ | 174.1/91 | $4 \times 10^{-7}$ |
| Symmetric bkg. ($A_b = 0$) | $-0.0471$ | 169.8/89 | $5 \times 10^{-7}$ |
| Single exponential | $-0.0480$ | 297.7/92 | $2 \times 10^{-23}$ |

TABLE VII. $A_{\mathrm{raw}}$ (posterior median) and goodness of fit of the nominal model (Gaussian signal, double exponential per charge) and of the alternative signal and background models, same data and window.

Table VII compares $A_{\mathrm{raw}}$ and the goodness of fit of the nominal model with those of the alternative signal and background models, on the same data.

Figure XX shows the mass distributions of (a) $B^+$ and (b) $B^-$ candidates with the nominal model, and the pulls below.

Figure XXI shows the posterior predictive distributions of (a) the binned $\chi^2$ and of the counting asymmetry of (b) the fit window and (c) the sidebands, against the observed values.

**6. Savage–Dickey Ratio** *In short: the Bayes factor of $A = 0$ is the ratio of the posterior and prior densities at $A = 0$, with no integral over the twelve parameters.*

*Proof.* Write $\theta$ for the parameters other than $A$. Let $H_1$ have $A$ free with the prior $\pi(A, \theta) = \pi(A)\,\pi(\theta)$, and let $H_0$ be $H_1$ with $A = 0$ and the same $\pi(\theta)$. The evidence of $H_0$ is

$$P(D \mid H_0) = \int \mathcal{L}(D \mid A = 0, \theta)\, \pi(\theta)\, d\theta,$$

and that of $H_1$ is $P(D \mid H_1) = \iint \mathcal{L}(D \mid A, \theta)\, \pi(A)\, \pi(\theta)\, dA\, d\theta$. The marginal posterior density of $A$ in $H_1$, evaluated at $A = 0$, is obtained by Bayes' theorem and by integrating over $\theta$:

$$\begin{aligned} p(A = 0 \mid D) &= \int \frac{\mathcal{L}(D \mid A = 0, \theta)\, \pi(A = 0)\, \pi(\theta)}{P(D \mid H_1)}\, d\theta \\ &= \pi(A = 0)\, \frac{P(D \mid H_0)}{P(D \mid H_1)}. \end{aligned}$$

Dividing by $\pi(A = 0)$ gives

$$BF_{01} = \frac{P(D \mid H_0)}{P(D \mid H_1)} = \frac{p(A = 0 \mid D)}{\pi(A = 0)}.$$

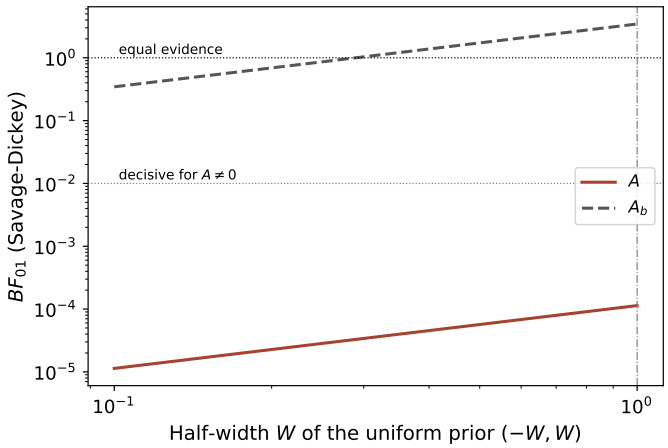



FIG. XVIII. $BF_{01}$ of $A = 0$ and of $A_b = 0$ as a function of the half-width $W$ of the uniform prior $(-W, W)$ (Gaussian approximation of the posterior, Savage-Dickey). The vertical line marks the nominal prior.

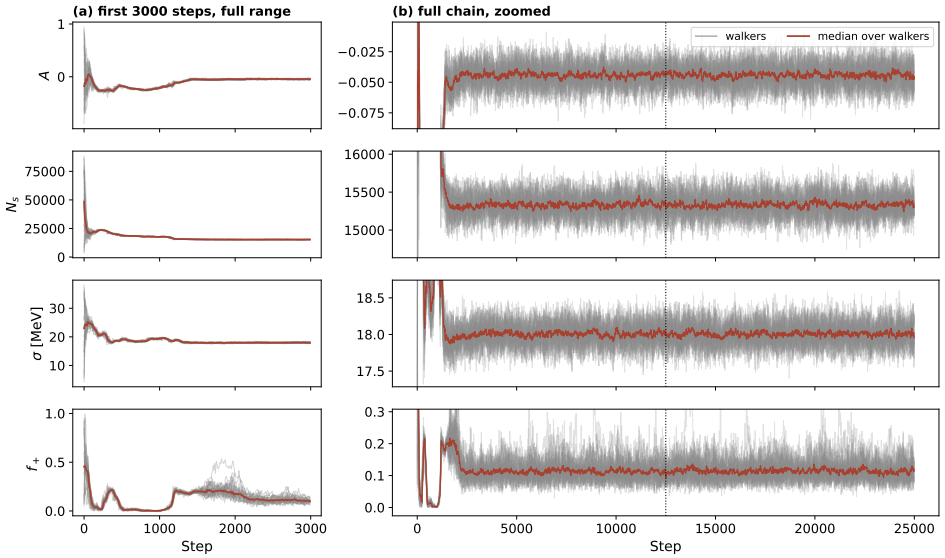



FIG. XIX. Trace plot of the nominal fit with the 32 walkers initialized uniformly over the prior box (one grey line per walker, every 10th step; colored: median over the walkers), for $A$, $N_s$, $\sigma$ and the steep-background fraction $f_+$. Left: first 3000 steps, full range; right: whole chain, zoomed on the posterior. The first half of the chain (dotted line) is discarded as burn-in.

It holds only if $A$ and $\theta$ are independent a priori and the other priors are shared by the two hypotheses; the priors must be proper (here $\pi(A = 0) = \frac{1}{2}$), otherwise the Bayes factor is not defined.

**7. Equal-Tailed and Highest-Posterior-Density Intervals** *In short: for $A$ and $A_b$ the two kinds of 68% interval coincide; they differ for the skewed parameters, whose highest-posterior-density interval is shifted towards the mode (Table VIII).*

Table VIII lists, for each parameter, the equal-tailed and highest-posterior-density 68% intervals and the skewness of the marginal posterior.

| Parameter | median | equal-tailed | HPD | skewness |
|---|---|---|---|---|
| $A$ | $-0.0442$ | $[-0.0529, -0.0358]$ | $[-0.0525, -0.0355]$ | $+0.01$ |
| $A_b$ | $-0.0400$ | $[-0.0575, -0.0224]$ | $[-0.0575, -0.0225]$ | $+0.01$ |
| $f_+$ | 0.114 | $[0.089, 0.145]$ | $[0.084, 0.138]$ | $+0.90$ |
| $f_-$ | 0.209 | $[0.174, 0.248]$ | $[0.168, 0.241]$ | $+0.33$ |
| $\lambda_1^+$ [MeV$^{-1}$] | 0.0350 | $[0.0281, 0.0431]$ | $[0.0274, 0.0423]$ | $+0.52$ |

TABLE VIII. Equal-tailed and highest-posterior-density 68% intervals and skewness of the marginal posteriors.

**8. Posterior Predictive Check with Statistics Not Fitted by the Model** *In short: the asymmetry in slices of the mass, which the model predicts, is reproduced ($p_B = 0.67$, Fig. XXIIb), and the difference between the $\chi^2$ of the two charges is at the edge of the replicas ($p_B = 0.08$, Fig. XXIIc).*

Figure XXII shows (a) the counting asymmetry in

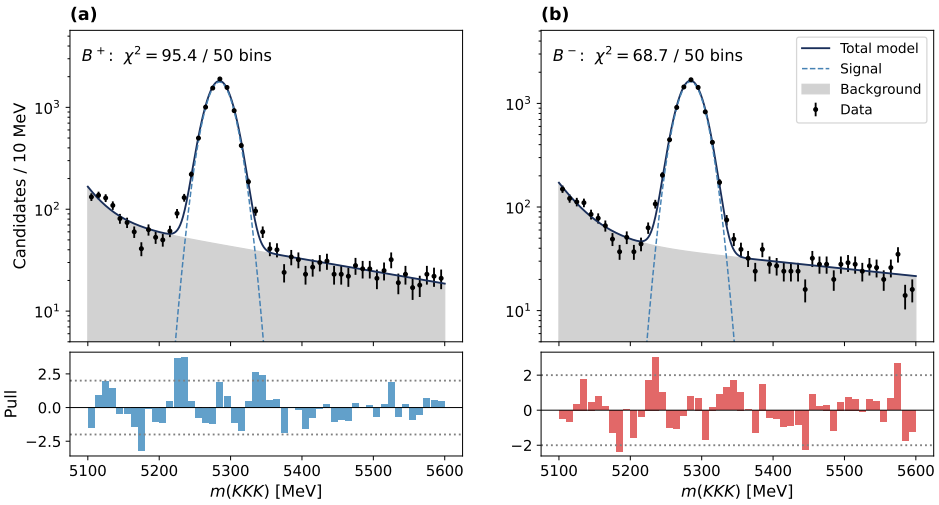



FIG. XX. Mass distributions of (a) $B^+$ and (b) $B^-$ candidates with the nominal model (posterior medians, logarithmic scale): total, signal and background components. Bottom: pulls, $(\text{obs} - \text{exp})/\sqrt{\text{exp}}$.

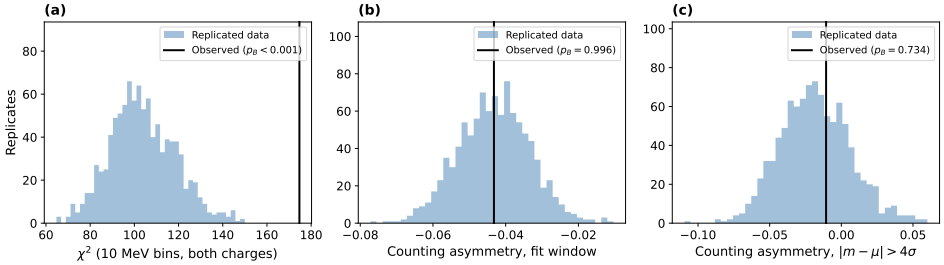



FIG. XXI. Posterior predictive check: distributions of three discrepancies over 1000 datasets replicated from posterior draws, compared with the observed values. (a) binned $\chi^2$; (b) and (c) counting asymmetry of the fit window and of the sidebands.

slices of 50 MeV of the mass, data and model, (b) the distribution of $T = \sum_s (A_s - \bar{A}_s)^2/\sigma_s^2$ over 1000 replicas with the observed value, and (c) the same for $\chi^2(B^+) - \chi^2(B^-)$.

**9. Robustness Check: Magnet Polarity** *In short: the posteriors of $A_{\text{raw}}$ for the magnet-down and magnet-up samples and for their combination agree within $0.7\sigma$ (Fig. XXIII).*

Figure XXIII shows the posterior of $A_{\text{raw}}$ for the magnet-down, magnet-up and combined samples.

**10. Toy-MC Coverage Validation** *In short: 200 toy-MC experiments give coverage $0.65 \pm 0.03$ (68%) and $0.93 \pm 0.02$ (95%) (Fig. XXIVb), consistent with nominal, with unbiased estimates and pulls of width 1.04 (Fig. XXIVa).*

Figure XXIV shows (a) the pulls of the posterior median and standard deviation of each replica against a standard normal, and (b) the fraction of replicas whose 68% and 95% credible intervals contain the injected value, with binomial uncertainties.

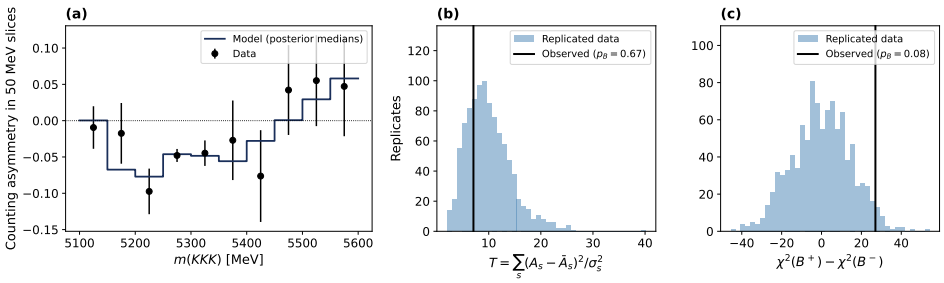



FIG. XXII. (a) Counting asymmetry in slices of 50 MeV of the mass, data and model. (b) Distribution of $T = \sum_s (A_s - \bar{A}_s)^2/\sigma_s^2$ over 1000 replicas and observed value. (c) The same for $\chi^2(B^+) - \chi^2(B^-)$.

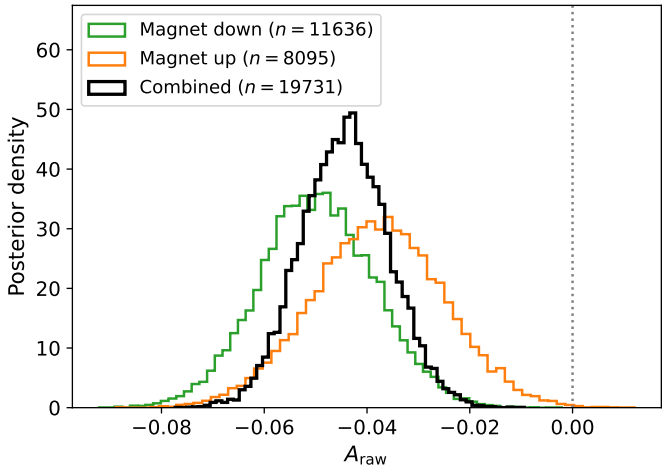



FIG. XXIII. Posterior of $A_{\text{raw}}$ for the magnet-down, magnet-up and combined samples.

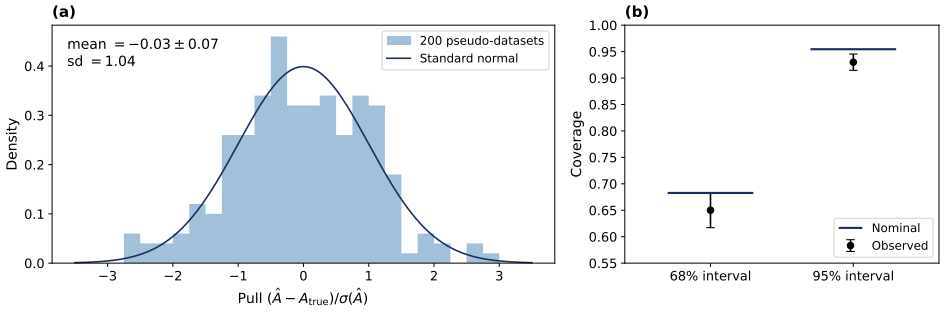



FIG. XXIV. Toy-MC validation, 200 toy-MC datasets. (a) Pulls $(\hat{A} - A_{\text{true}})/\sigma(\hat{A})$, with $\hat{A}$ and $\sigma(\hat{A})$ the posterior median and standard deviation of each replica, compared with a standard normal. (b) Fraction of replicas whose 68% and 95% credible intervals contain the injected value, with binomial uncertainties, and the nominal coverage.

## Appendix E: Validation of the Estimator and of the Intervals

*In short: the size and power of the test of $A = 0$, the asymptotic behavior of the estimator and the goodness of fit as a function of the binning.*

**1. Size and Power of the Test of $A = 0$** *In short: the statistic $q(0)$ follows a $\chi^2$ distribution with one degree of freedom (Fig. XXV), so the test has the nominal size; an asymmetry of $-0.02$ would be rejected at $3\sigma$ in one toy-MC experiment out of four (Fig. VIIc).*

Figure XXV shows, for toy-MC experiments, (a) the fraction with $q(0)$ above the observed value for $A_{\text{true}} = 0$ against the $\chi^2_1$ distribution, (b) the $p$-value of $t(A_{\text{true}})$ against a uniform distribution, at the fitted value and pooled over the injected values, and (c) the distribution of $t(A_{\text{true}})$ at the fitted value against $\chi^2_1$.

**2. Asymptotic Behavior of the Estimator** *In short: with the number of candidates the bias tends to zero, the spread scales as $N^{-1/2}$ and the distribution of $\hat{A}$ becomes Gaussian (Table IX), as the central limit theorem and the asymptotic properties of the maximum-likelihood estimate predict.*

Table IX gives, for $N$ from 50 to 2000 candidates, the bias, the spread and its product with $\sqrt{N}$, the skewness, the excess kurtosis and the Shapiro $p$-value of $\hat{A}$ in 300 toy-MC datasets.

**3. Goodness of Fit and the Binning** *In short: the binned $\chi^2$ of toy-MC experiments follows its asymptotic distribution, so its p-values are accurate; the misfit of the data is 5.4 standard deviations at 10 MeV bins (Fig. XXVIa) and 2.0 at 2.5 MeV (Fig. XXVIb), because it lives on the scale of the flanks of the peak.*

Figure XXVI shows the binned $\chi^2$ of 1000 toy-MC experiments against the $\chi^2$ distribution with the corresponding degrees of freedom, and the value of the data, for (a) $2 \times 50$ bins of 10 MeV and (b) $2 \times 200$ bins of 2.5 MeV.

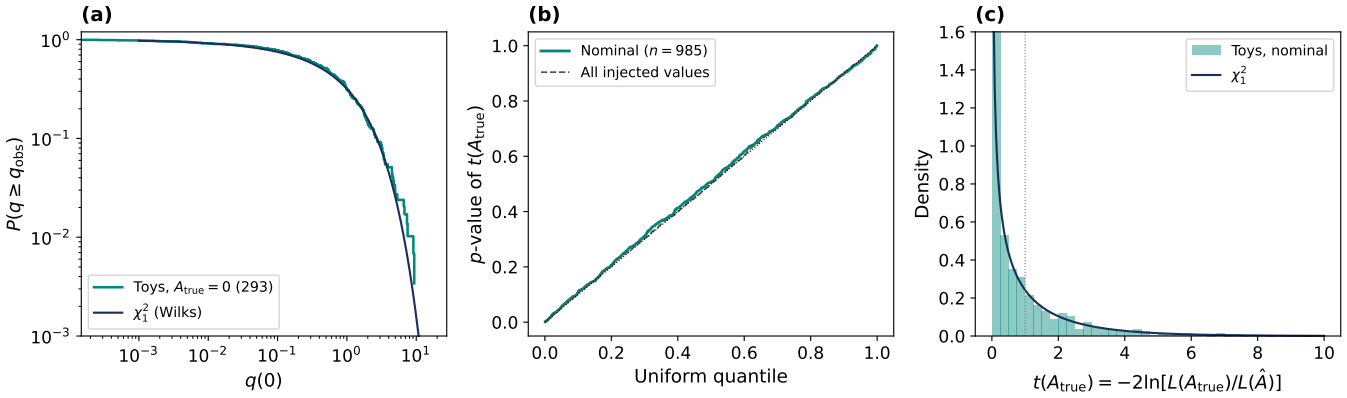



FIG. XXV. Toy-MC experiments, validity of Wilks' theorem. (a) Fraction of experiments with $q(0)$ above the observed value, for $A_{\rm true} = 0$, compared with the $\chi^2_1$ distribution. (b) $p$-value of $t(A_{\rm true})$ against a uniform distribution, at the fitted value and pooled over the injected values. (c) Distribution of $t(A_{\rm true})$ at the fitted value and $\chi^2_1$.

| $N$ | bias | sd($\hat{A}$) | sd$\cdot\sqrt{N}$ | skewness | ex. kurtosis | Shapiro $p$ |
|---|---|---|---|---|---|---|
| 50 | −0.0085 | 0.1611 | 1.14 | −0.15 | +0.64 | 0.03 |
| 150 | −0.0051 | 0.0920 | 1.13 | +0.16 | +0.31 | 0.59 |
| 500 | −0.0022 | 0.0501 | 1.12 | −0.08 | +0.38 | 0.52 |
| 2000 | +0.0009 | 0.0225 | 1.01 | −0.20 | −0.20 | 0.32 |

TABLE IX. Distribution of the maximum-likelihood estimate of $A$ in 300 toy-MC datasets of $N$ candidates.

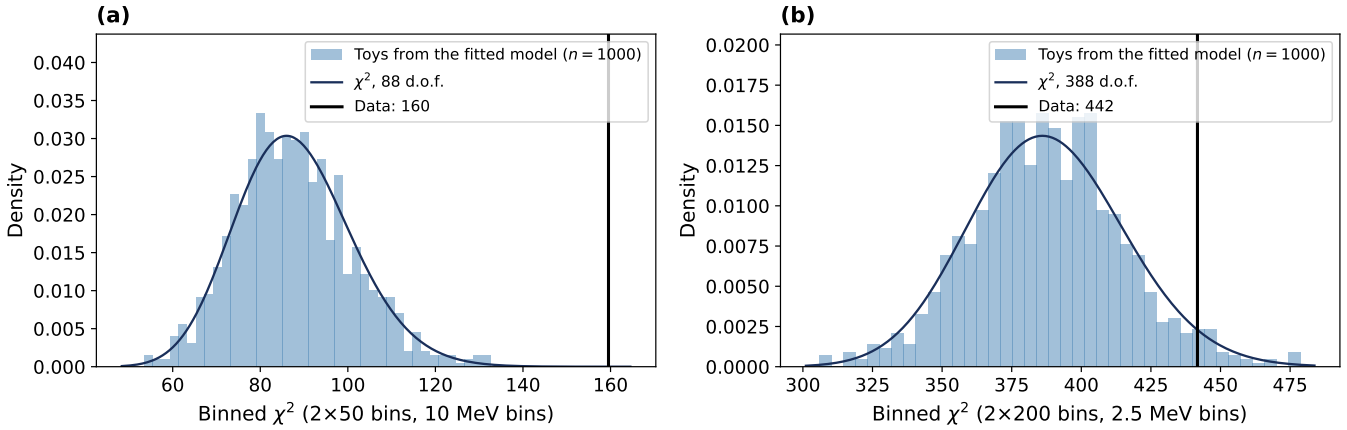



FIG. XXVI. Binned $\chi^2$ of 1000 toy-MC experiments from the fitted model, $\chi^2$ distribution with the corresponding degrees of freedom, and value of the data: (a) $2 \times 50$ bins of 10 MeV, (b) $2 \times 200$ bins of 2.5 MeV.

## Appendix F: Machine-Learning Analysis

*In short: classifiers trained on weak labels from the mass spectrum (there are no true labels) reach a test AUC of* 0.86*; a cut on their score leaves $A_{\mathrm{raw}}$ unchanged, and an adaptive partition of the Dalitz plane shows the same pattern as the fixed binning.*

**1. Model Comparison** *In short: six classifiers; the tree ensembles and the neural network reach test AUC* 0.857–0.864*, about* 0.09 *above the linear models and indistinguishable from each other within the test-set uncertainty (*0.012*; Table X, Fig. XXVIIa); gradient boosting, best in cross-validation, is the reference.*

Table X gives the cross-validated AUC, the test-set AUC with its bootstrap uncertainty and the Brier score of each model.

Figure XXVII shows (a) the ROC curves and (b) the significance improvement of a cut on the score against the signal-like efficiency.

Figure XXVIII compares (a) the Random Forest feature importance with (b) the magnitude of the Lasso coefficients.

**2. Background: Regularization, Bias-Variance and Loss Functions** *In short: cross-entropy is a Bernoulli negative log-likelihood, L1 and L2 penalties shrink the coefficients, the test error splits into bias, variance and noise, and class weights rescale the score by a known factor.*

**3. Regularization, Bias-Variance, and Calibration** *In short: the linear model does not overfit (Fig. XXIX), gradient boosting overfits beyond ∼150 stages without losing test AUC (Fig. XXX), and class weights bias the probabilities, which an isotonic recalibration corrects (Brier score* 0.122 → 0.062*, Fig. XXXI) without changing the ranking.*

Figure XXIX shows the cross-validated AUC of the logistic regression against the inverse regularization strength, for the $L_1$ and $L_2$ penalties.

Figure XXX shows the accuracy (a) and the AUC (b) of gradient boosting on the training and test sets against the number of stages.

Figure XXXI is the reliability diagram of the Random Forest, raw and recalibrated, with the distribution of the raw score below.

**4. Uncertainty Quantification** *In short: the three estimators give mean per-candidate uncertainties of* 0.178 *(Random Forest),* 0.095 *(MC dropout) and* 0.050 *(deep ensemble; Fig. XXXIIb); they measure different sources of variability, and only the Random Forest spread visibly decreases with the training-set size (Fig. XXXIIc).*

Figure XXXII shows (a) the spread across the Random Forest trees against the mean tree score for each weak-label class, (b) the distributions of the per-candidate uncertainty of the three estimators and (c) the mean Random Forest spread against the training-set size.

**5. Stability of $A_{\mathrm{raw}}$ under ML Selection** *In short: a cut on the score at* 0.5 *keeps* 92% *of the signal and* 36% *of the background and leaves $A_{\mathrm{raw}}$ within its expected statistical spread with both fits and with counting (Table XI, Fig. XXXIII); it does not improve the precision and cannot remove the misidentified background (Fig. XXXIV).*

Table XI gives $A_{\mathrm{raw}}$ on the full window and after the cuts at 0.5 and 0.73, by the three methods, with the expected shift. The tighter cut at the maximum of the significance improvement (score $> 0.73$, $\varepsilon_S = 0.66$, $\varepsilon_B = 0.13$, Fig. XXVIIb) gives shifts of $+0.004$ to $+0.005$, 0.9 times their expected spread, and $A \neq 0$ is still $4.1\sigma$; this cut is not an optimum for the measurement, since $\varepsilon_S/\sqrt{\varepsilon_B}$ holds for a signal small against the background, while here the precision follows $S/\sqrt{S+B}$ and worsens from 0.0085 to 0.0097.

Figure XXXIII shows the counting asymmetry after a cut on the gradient-boosting score against the threshold, with the value of the full window and the band of the expected statistical spread of a nested subsample.

Figure XXXIV shows (a) the mass spectrum of all candidates and of those passing the cut, for the 20-feature classifier and for one trained without the kaon momentum components, and (b) the fraction passing the cut.

Figure XXXV shows, in a high-background sample (`ProbK` $> 0.3$), (a) the statistical uncertainty and (b) the value of $A_{\mathrm{raw}}$ against the signal efficiency, for tighter identification cuts and for cuts on the score of a classifier retrained on this sample.

**6. Background: Clustering and Uncertainty Estimators** *In short: K-means converges to a local minimum of the inertia, the adjusted Rand index measures the stability of a partition, and the spread of an ensemble estimates the epistemic uncertainty.*

**7. K-means on the Dalitz Plot** *In short: for $K = 8$, fixed in advance, two clusters deviate by more than $3\sigma$ ($\chi^2 = 46.4$ for 7 degrees of freedom; Fig. IX), while the mean silhouette is maximal at $K = 3$ (Fig. XXXVI).*

Figure XXXVI shows the inertia (a) and the mean silhouette (b) of the $K$-means partition of the Dalitz plane for $K = 2$–12, with the $K = 8$ fixed in advance and the maximum of the silhouette marked.

**8. Hyperparameter Optimization** *In short: the best configurations improve the test AUC by at most* 0.005*, less than its uncertainty (Tables XII and XIII), and the cross-validated maximum overestimates the test AUC (winner's curse; Fig. XXXVII).*

Table XII gives the search for the tree ensembles, with the test AUC of the best configuration against that of the default.

Table XIII lists the five best feed-forward configurations, within 0.0007 of each other.

Figure XXXVII shows the cross-validated AUC of the 168 feed-forward configurations for each of the seven architectures.

| Model | CV AUC | Test AUC | Brier |
|---|---|---|---|
| Logistic Regression | $0.769 \pm 0.016$ | $0.771 \pm 0.013$ | 0.197 |
| Logistic Regression (L1) | – | $0.771 \pm 0.013$ | 0.198 |
| Decision Tree | $0.816 \pm 0.011$ | $0.824 \pm 0.013$ | 0.133 |
| Random Forest | $0.863 \pm 0.017$ | $0.860 \pm 0.012$ | 0.122 |
| Gradient Boosting | $\mathbf{0.866 \pm 0.016}$ | $0.864 \pm 0.012$ | 0.116 |
| FFNN | – | $0.857 \pm 0.012$ | 0.134 |

TABLE X. Cross-validated AUC (mean and standard deviation over the folds), test-set AUC (with bootstrap uncertainty) and test-set Brier score of the raw scores. Gradient boosting, best in cross-validation, is the reference model. The FFNN is validated on the validation set instead of by cross-validation.

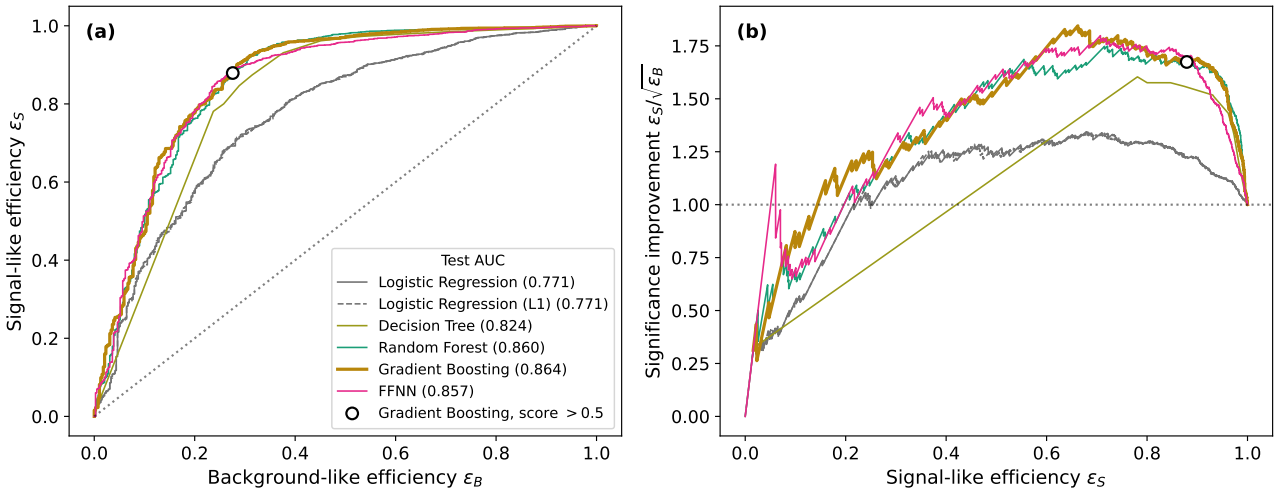



FIG. XXVII. Test-set performance of the classifiers: (a) ROC curves, with the test AUC of each model in the legend (Table X); (b) significance improvement $\varepsilon_S/\sqrt{\varepsilon_B}$ of a cut on the score as a function of the signal-like efficiency, where 1 is no selection and $\varepsilon_{S,B}$ are the fractions of signal-like and background-like candidates kept. The reference model is drawn thicker, and the two logistic regressions overlap. The circle marks the cut on the reference score used in Appendix F.5.

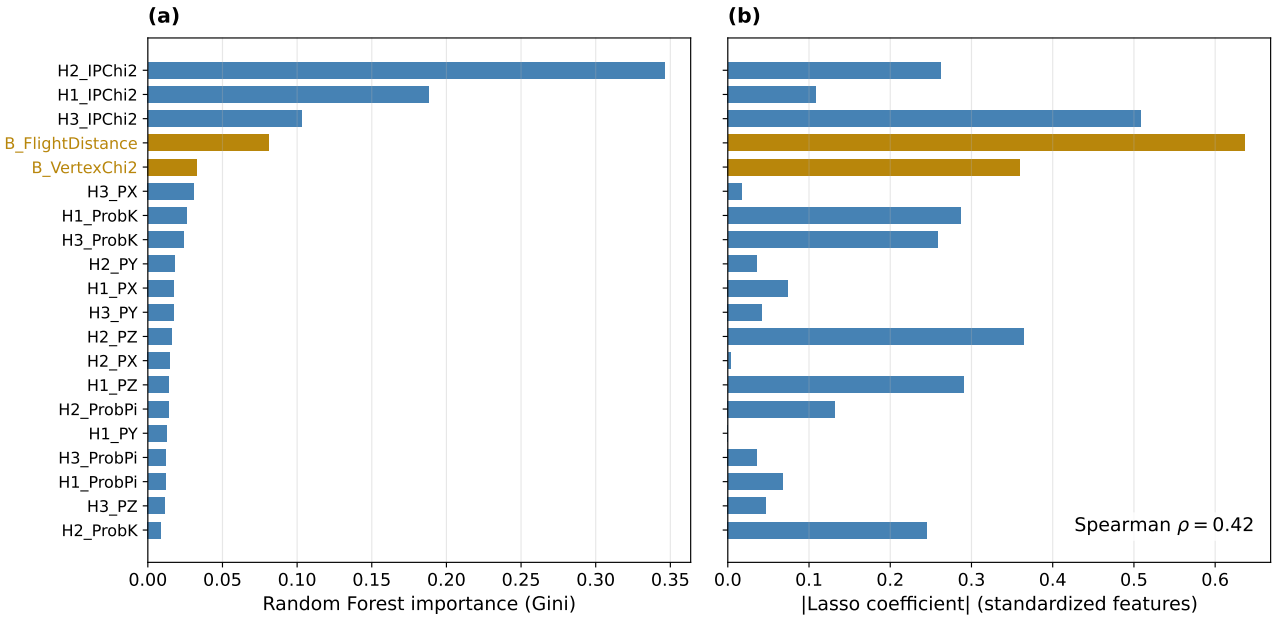



FIG. XXVIII. Random Forest feature importance (Gini, a) and magnitude of the Lasso coefficient of the standardized feature (b) for the 20 features, both ordered by Random Forest importance; the two $B$-candidate variables are highlighted.

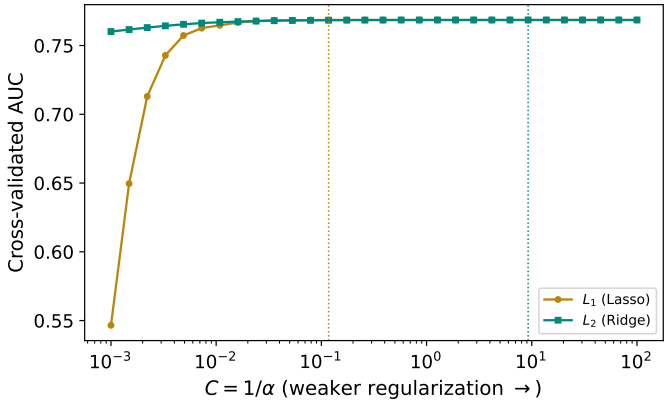



FIG. XXIX. Cross-validated AUC of the logistic regression against the inverse regularization strength $C = 1/\alpha$ (30 log-spaced values), for the $L_1$ (Lasso) and $L_2$ (Ridge) penalties; the dotted lines mark the optimum of each.

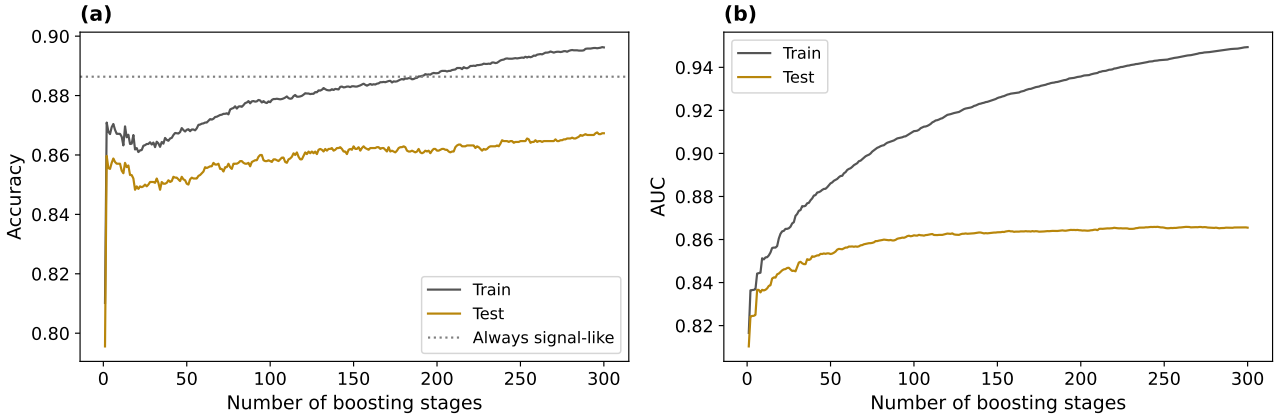



FIG. XXX. Gradient boosting: (a) accuracy at the 0.5 threshold and (b) AUC on the training and test sets as a function of the number of boosting stages. The dotted line is the accuracy of always predicting the signal-like class.

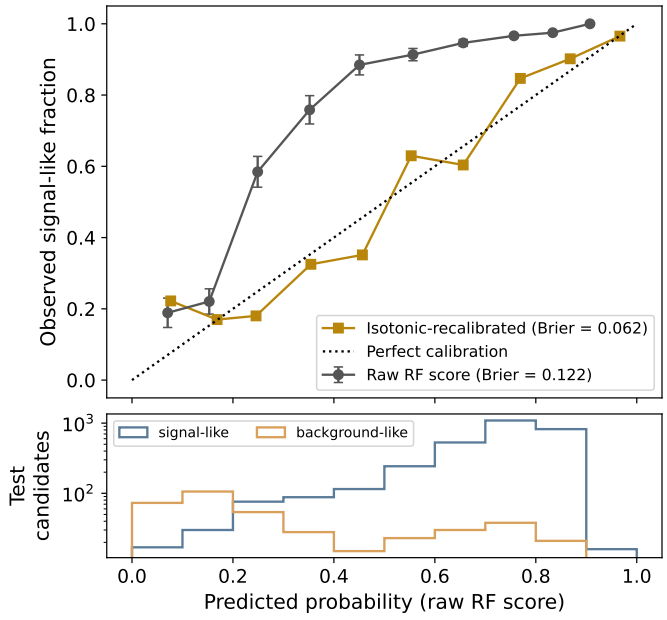



FIG. XXXI. Reliability diagram of the Random Forest on the test set: observed fraction of signal-like candidates vs. predicted probability, for the raw score (with binomial uncertainties) and for the isotonic-recalibrated one. Bottom: distribution of the raw score for the two weak-label classes; the low-probability bins contain few candidates.

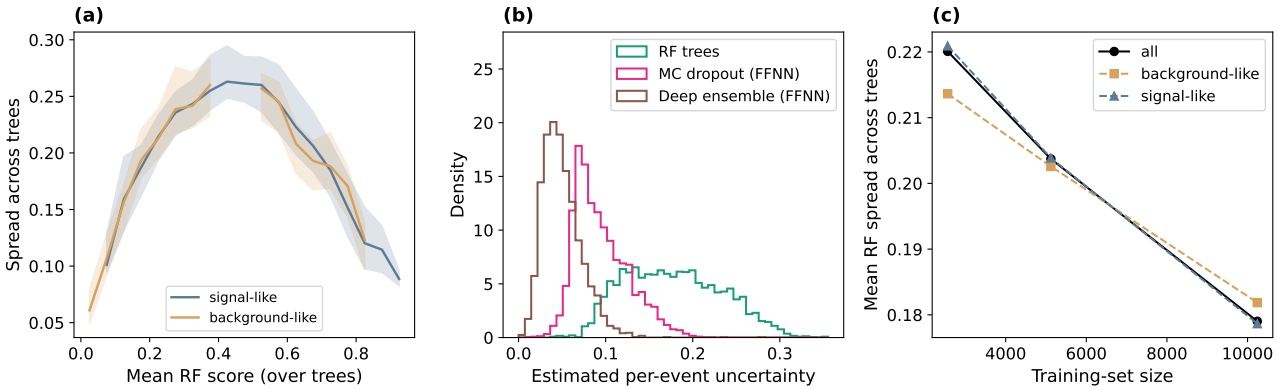



FIG. XXXII. (a) Spread across the Random Forest trees vs. mean tree score, median and 16–84% band of the test candidates of each weak-label class. (b) Distributions of the per-candidate uncertainty from the Random Forest trees, MC dropout and the deep ensemble. (c) Mean Random Forest spread vs. training-set size, for all, background-like and signal-like test candidates.

| Method | Full sample | Score $> 0.5$ | Shift (expected) | Score $> 0.73$ | Shift (expected) |
|---|---|---|---|---|---|
| Bayesian fit | $-0.0443 \pm 0.0085$ | $-0.0451 \pm 0.0087$ | $-0.0008\,(0.0018)$ | $-0.0404 \pm 0.0097$ | $+0.0039\,(0.0047)$ |
| Frequentist fit | $-0.0441 \pm 0.0085$ | $-0.0447 \pm 0.0086$ | $-0.0006\,(0.0012)$ | $-0.0401 \pm 0.0097$ | $+0.0040\,(0.0048)$ |
| Counting | $-0.0432 \pm 0.0071$ | $-0.0446 \pm 0.0080$ | $-0.0014\,(0.0036)$ | $-0.0381 \pm 0.0093$ | $+0.0051\,(0.0059)$ |

TABLE XI. $A_{\mathrm{raw}}$ on the full fit window (19,731 candidates) and after the cuts on the gradient-boosting score at 0.5 (15,709 candidates) and at 0.73 (11,658), the maximum of the significance improvement of Fig. XXVII, by the three methods. The expected shift is the statistical spread of a nested subsample, $\sqrt{\sigma^2_{\mathrm{sub}} - \sigma^2_{\mathrm{full}}}$. Counting includes the background.

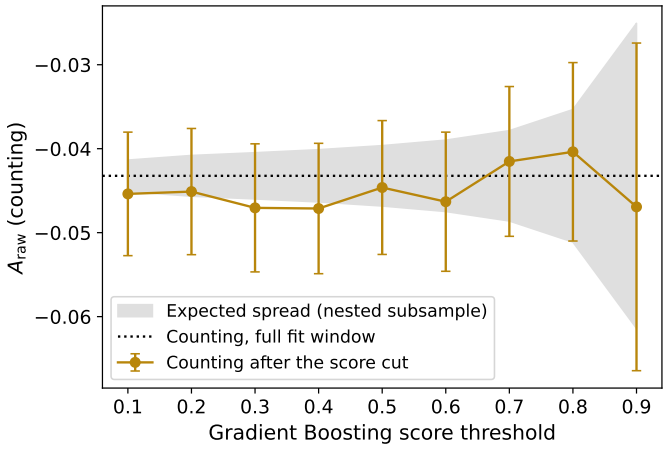



FIG. XXXIII. Counting asymmetry after a cut on the gradient-boosting score, vs. threshold, with its binomial uncertainty. Dotted line: counting asymmetry of the full fit window; band: expected statistical spread of a nested subsample around it, $\pm\sqrt{\sigma^2_{\mathrm{sub}} - \sigma^2_{\mathrm{full}}}$.

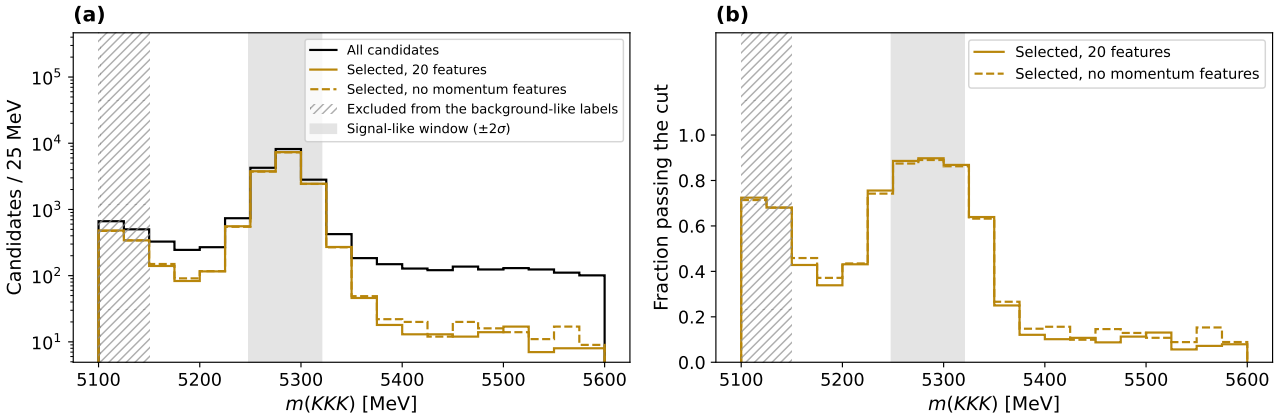



FIG. XXXIV. (a) Mass spectrum of all candidates and of those with gradient-boosting score $> 0.5$, for the 20-feature classifier and for one trained without the kaon momentum components (25 MeV bins). (b) Fraction of candidates passing the cut. Hatched: the region below 5150 MeV excluded from the background-like labels; grey: the signal-like window.

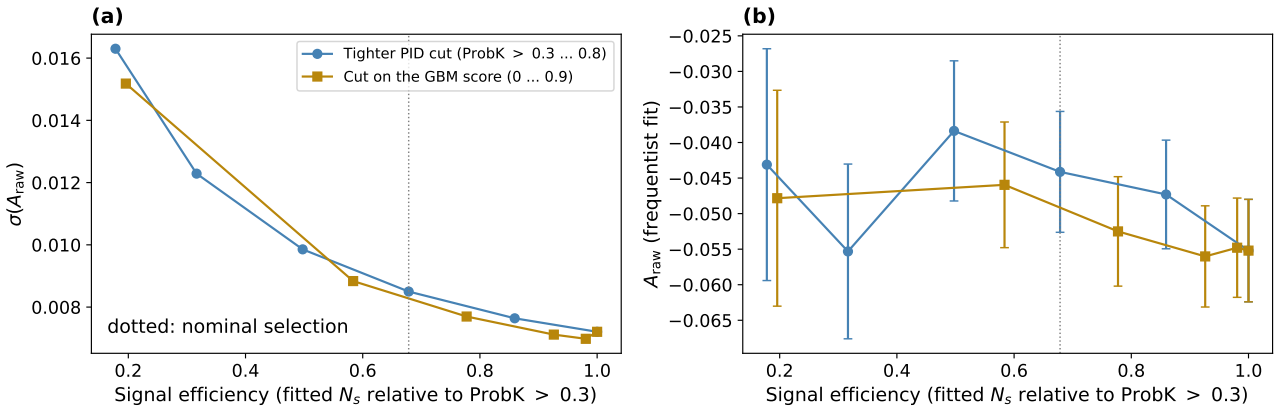



FIG. XXXV. High-background sample (`ProbK` $> 0.3$): (a) statistical uncertainty and (b) value of $A_{\mathrm{raw}}$ from the frequentist fit vs signal efficiency, for tighter identification cuts (`ProbK` $> 0.3, \ldots, 0.8$) and for cuts on the score of the classifier retrained on this sample $(0, \ldots, 0.9)$. The dotted line marks the nominal selection (`ProbK` $> 0.5$). The samples are nested, so their uncertainties are correlated.

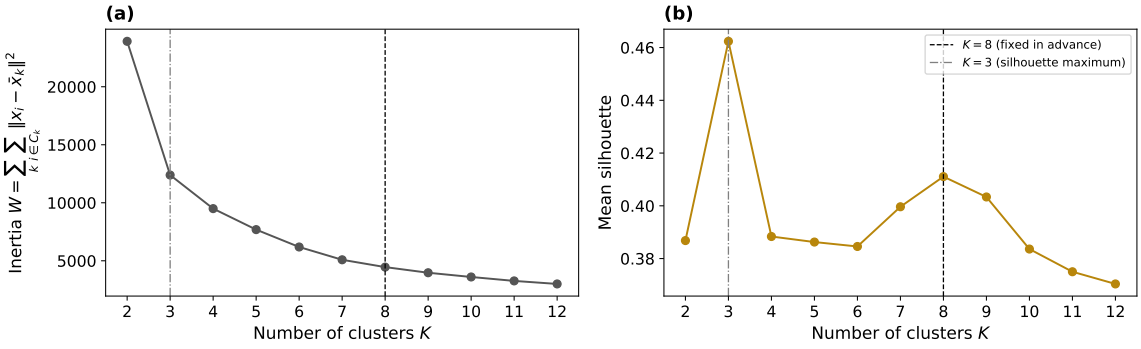



FIG. XXXVI. (a) Inertia and (b) mean silhouette of the $K$-means partition of the Dalitz plane for $K = 2$–12; the dashed line marks the $K = 8$ fixed in advance and the dash-dotted line the maximum of the silhouette, $K = 3$.

| Model | Grid | Best configuration | CV AUC / Test AUC (default) |
|---|---|---|---|
| Random Forest | `n_estimators` 200–1200, `max_depth` 6–12 or none, `min_samples_leaf` 5–40 | 400, 12, 5 | 0.8706 / 0.8656 (0.8604) |
| Gradient Boosting | `n_estimators` 100–500, `max_depth` 1–5, `learning_rate` 0.01–0.1 | 300, 2, 0.1 | 0.8684 / 0.8650 (0.8643) |

TABLE XII. Hyperparameter search for the tree ensembles (120 and 100 configurations, 5-fold cross-validation repeated 10 times, balanced class weights). The test AUC of the best configuration is compared with that of the default of Table X (in parentheses); the bootstrap uncertainty of each test AUC is 0.012.

| Architecture | Layers | Dropout | Learning rate | CV AUC |
|---|---|---|---|---|
| `deep_3layer` (best) | [64, 32, 16] | 0.2 | 0.003 | $0.8585 \pm 0.0174$ |
| `wide_2layer` | [64, 32] | 0.3 | 0.003 | $0.8582 \pm 0.0175$ |
| `deep_3layer` | [64, 32, 16] | 0.3 | 0.003 | $0.8580 \pm 0.0187$ |
| `wide_2layer` | [64, 32] | 0.3 | 0.005 | $0.8579 \pm 0.0184$ |
| `wide_2layer` | [64, 32] | 0.4 | 0.005 | $0.8579 \pm 0.0187$ |

TABLE XIII. Top five of the 168 FFNN configurations (7 architectures from one hidden layer of 8 units to five layers [128, 64, 32, 16, 8], dropout 0.1–0.4, learning rates 0.0002–0.005; 5-fold cross-validation repeated twice). The five are within 0.0007 of each other, far below the fold-to-fold spread: the optimum is a plateau. The best configuration has test AUC 0.8606, against 0.8568 for the default architecture (±0.005 over initializations).

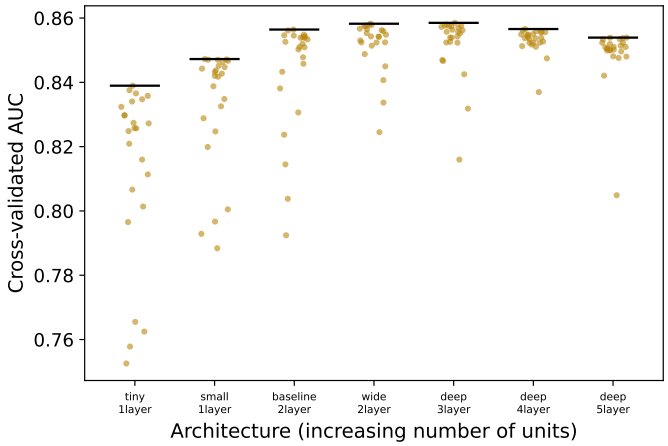



FIG. XXXVII. Cross-validated AUC of the 168 FFNN configurations (dots) for each of the seven architectures, ordered by number of units; the black segment marks the best configuration of each architecture.

### Appendix G: Local Asymmetry in the Dalitz Plane

*In short: the homogeneity test in other samples and in the two magnet polarities, the permutation test and the $\phi(1020)$ band.*

**1. Homogeneity Test: Signal Window, Vetoes, Grids and Magnet Split** *In short: the homogeneity test holds in the signal window, with the charm vetoes and for other grids, and the structure is the same in the two magnet polarities ($\chi^2 = 23.0$ for 29 bins).*

**2. Asymmetry in the $\phi(1020)$ Band and in the Significant Bins** *In short: the band of the $\phi(1020)$ has no asymmetry, the negative asymmetry is outside it and larger when the $\phi$ candidates are removed (Fig. XXXVIIIb); in the region of the published analysis the $B^-$ peak is visibly smaller (Fig. XXXIXb).*

Figure XXXVIII shows (a) the distribution of $m_{\rm lo}(K^+K^-)$ around the $\phi(1020)$ with the band used, and (b) the counting asymmetry of all candidates, of the $\phi$ band, of the region outside it and of the two significant bins with and without the $\phi$ candidates.

Figure XXXIX shows (a) the ordered Dalitz plot with the squared masses of the $\phi(1020)$, $D^0$, $J/\psi$ and $\chi_{c0}$ and the region of the published analysis, and (b) the mass spectra of $B^+$ and $B^-$ candidates in that region.

**3. Distribution of the Homogeneity Statistic under Charge Permutation** *In short: shuffling the charges reproduces the $\chi^2$ distribution of a uniform asymmetry, and the observed value lies far beyond all 2000 permutations (Fig. XL).*

Figure XL shows the homogeneity statistic $\chi^2 = \sum_i S_i^2$ of the $8 \times 8$ grid for 2000 random permutations of the charges, the $\chi^2$ distribution with 31 degrees of freedom and the observed value.

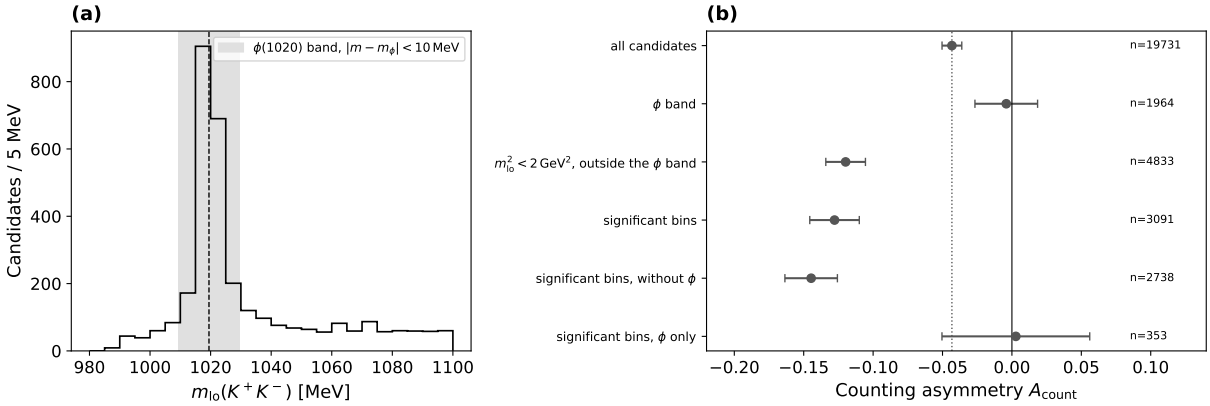



FIG. XXXVIII. (a) Distribution of $m_{\rm lo}(K^+K^-)$ around the $\phi(1020)$ (5 MeV bins) and band used (shaded). (b) Counting asymmetry, with binomial uncertainty and number of candidates, of all candidates (dotted line), of the $\phi$ band, of the region $m^2_{\rm lo} < 2$ GeV$^2$ outside the band, and of the two significant bins of Fig. XI, with and without the candidates of the $\phi$ band, and of those candidates alone.

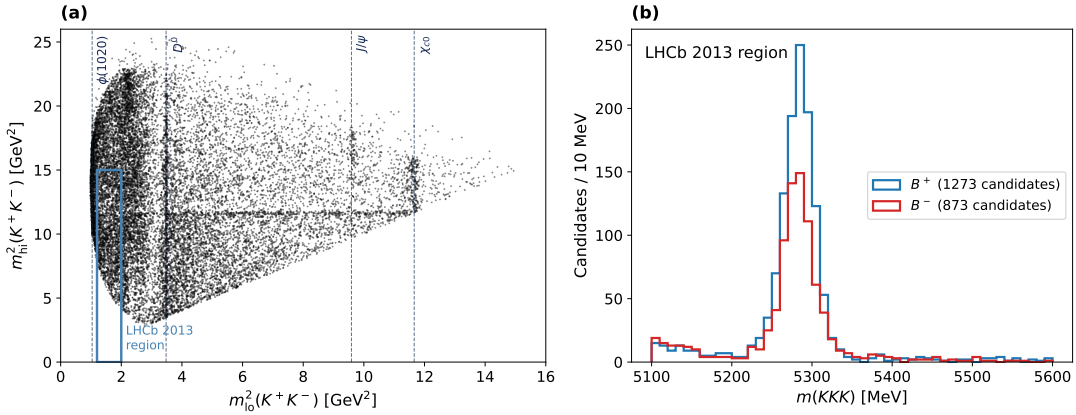



FIG. XXXIX. (a) Ordered Dalitz plot of the candidates in the fit window, with the squared masses of the $\phi(1020)$, $D^0$, $J/\psi$ and $\chi_{c0}$ (dashed) and the region of the published analysis of the same data [5] (box). (b) Mass spectra of $B^+$ and $B^-$ candidates in that region, same binning: the $B^-$ peak is visibly smaller.

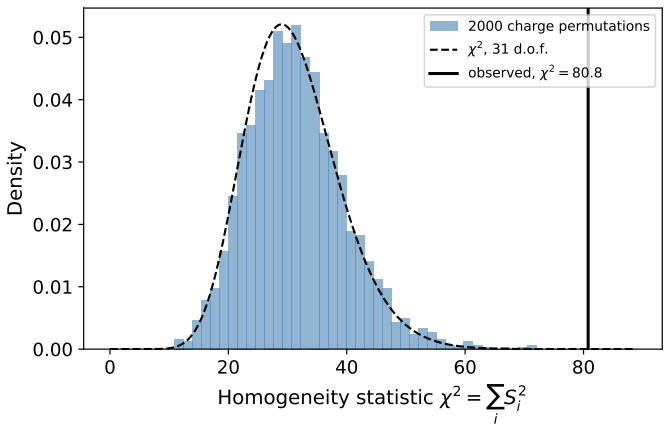



FIG. XL. Homogeneity statistic $\chi^2 = \sum_i S_i^2$ of the $8 \times 8$ Dalitz grid for 2000 random permutations of the charges (histogram), $\chi^2$ distribution with 31 degrees of freedom (dashed) and observed value (line).

### Appendix H: Systematic Uncertainties

*In short: every fit of the nominal model and of its variants with both methods, the validation of the combination rule on nested samples and the stability scans of the frequentist fit.*

**1. All Fits of the Nominal Model and of Its Variants** *In short: the Bayesian and frequentist results of every variant and cross-check side by side (Table XIV), and the shift of each variant with its expected spread $\sigma_B$ (Table XV).*

Table XIV lists, for every variant, the number of candidates, the Bayesian median and 68% credible interval of $A_{\mathrm{raw}}$, the binned goodness of fit and the frequentist estimate with its Hesse uncertainty.

Table XV gives the shift of each variant, the expected spread $\sigma_B$ for the variants that change the sample, and the uncertainty of each group.

**2. Validation of the Rule for Nested Samples** *In short: the spread of the shift of $\hat{A}$ between a sample and its tighter-identification subsample is the one given by the quadrature difference of the uncertainties (Fig. XLI).*

Figure XLI shows the shift of $\hat{A}$ between the nominal sample and a tighter selection, divided by the median $\sigma_B$, in 300 bootstrap replicas, for (a) `ProbK` $> 0.6$ and (b) `ProbPi` $< 0.4$, with a standard normal distribution.

**3. Stability Scans of the Frequentist Fit: Numbers** *In short: the largest shifts are* 0.0048 *over the grid of window edges (Fig. XLIIa) and* 0.0126 *over the grid of the two PID cuts (Fig. XLIIc), whose samples are smaller (mean uncertainty* 0.0112*); the signal region moves $A_{\mathrm{raw}}$ by $6 \times 10^{-5}$ (Fig. XLIIb).*

Figure XLII shows the shift of $A_{\mathrm{raw}}$ with respect to the nominal fit over three grids: (a) the lower and upper edge of the fit window, (b) the lower and upper edge of the signal region, and (c) the lower cut on `ProbK` and the upper cut on `ProbPi`.

| Fit | $n$ | $A_{\text{raw}}$ Bayesian (median) | 68% CI | $\chi^2$/ndof | $A_{\text{raw}}$ frequentist |
|---|---|---|---|---|---|
| Nominal | 19731 | $-0.0443$ | $[-0.0528, -0.0358]$ | 164.1/88 | $-0.0441 \pm 0.0085$ |
| *Systematic variants (Sec. IX)* | | | | | |
| Window $[5050, 5600]$ | 21164 | $-0.0446$ | $[-0.0531, -0.0362]$ | 217.4/98 | $-0.0445 \pm 0.0085$ |
| Window $[5150, 5600]$ | 18566 | $-0.0444$ | $[-0.0528, -0.0359]$ | 141.5/78 | $-0.0446 \pm 0.0088^{\dagger}$ |
| Window $[5100, 5450]$ | 19004 | $-0.0440$ | $[-0.0524, -0.0357]$ | 136.9/58 | $-0.0442 \pm 0.0085$ |
| Window $[5100, 5700]$ | 20191 | $-0.0456$ | $[-0.0540, -0.0373]$ | 200.2/108 | $-0.0455 \pm 0.0085$ |
| Single exponential | 19731 | $-0.0480$ | $[-0.0564, -0.0395]$ | 297.7/92 | $-0.0481 \pm 0.0085$ |
| Background shape shared | 19731 | $-0.0480$ | $[-0.0564, -0.0397]$ | 174.1/91 | $-0.0481 \pm 0.0083$ |
| Symmetric background | 19731 | $-0.0471$ | $[-0.0552, -0.0388]$ | 169.8/89 | $-0.0472 \pm 0.0084$ |
| Crystal Ball signal | 19731 | $-0.0445$ | $[-0.0529, -0.0363]$ | 131.0/86 | $-0.0447 \pm 0.0085$ |
| $\mu, \sigma$ per charge | 19731 | $-0.0443$ | $[-0.0527, -0.0356]$ | 164.2/86 | $-0.0442 \pm 0.0087$ |
| `ProbK` $> 0.4$ | 26553 | $-0.0474$ | $[-0.0550, -0.0399]$ | 175.6/88 | $-0.0473 \pm 0.0076$ |
| `ProbK` $> 0.6$ | 13981 | $-0.0385$ | $[-0.0482, -0.0289]$ | 145.0/88 | $-0.0384 \pm 0.0099$ |
| `ProbPi` $< 0.4$ | 17984 | $-0.0421$ | $[-0.0506, -0.0333]$ | 150.1/88 | $-0.0421 \pm 0.0088$ |
| `ProbPi` $< 0.6$ | 20953 | $-0.0446$ | $[-0.0529, -0.0366]$ | 159.0/88 | $-0.0448 \pm 0.0083$ |
| *Cross-checks (Secs. V B, D.9)* | | | | | |
| Dispersed start, run 1 | 19731 | $-0.0446$ | $[-0.0529, -0.0360]$ | 164.1/88 | – |
| Dispersed start, run 2 | 19731 | $-0.0442$ | $[-0.0528, -0.0356]$ | 164.1/88 | – |
| Magnet down | 11636 | $-0.0497$ | $[-0.0607, -0.0386]$ | 153.1/88 | – |
| Magnet up | 8095 | $-0.0378$ | $[-0.0507, -0.0251]$ | 150.3/88 | – |
| Narrow prior, $A \in (-0.3, 0.3)$ | 19731 | $-0.0443$ | $[-0.0528, -0.0358]$ | 164.1/88 | – |
| $D^0$ veto | 18744 | $-0.0496$ | $[-0.0584, -0.0409]$ | 159.1/88 | $-0.0493 \pm 0.0087$ |
| $D^0$, $J/\psi$, $\chi_{c0}$ vetoes | 16367 | $-0.0423$ | $[-0.0516, -0.0329]$ | 140.1/88 | $-0.0421 \pm 0.0094$ |

TABLE XIV. All fits of the nominal model and of its variants: number of candidates, Bayesian posterior median and 68% credible interval of $A_{\text{raw}}$, binned goodness of fit (10 MeV bins), and frequentist estimate with its Hesse uncertainty (Sec. IV; fitted on the same candidates; the cross-checks of the Bayesian fits that have no frequentist counterpart in this work are marked –). $^{\dagger}$ Uncertainty scaled from the number of events, since the Hesse matrix of that fit is not accurate. Every fit uses 32 walkers and 25,000 steps.

| Group | Variant | Bayesian | Frequentist |
|---|---|---|---|
| Window | $[5050, 5600]$ | $-0.0003$ (0.0009) | $-0.0004$ (0.0005) |
| | $[5150, 5600]$ | $-0.0001$ (0.0007) | $-0.0005$ (0.0021) |
| | $[5100, 5450]$ | $+0.0002$ (0.0009) | $-0.0001$ (0.0003) |
| | $[5100, 5700]$ | $-0.0014$ (0.0010) | $-0.0014$ (0.0001) |
| | *group uncertainty* | *0.0014* | *0.0014* |
| Background | single exponential (rejected) | $-0.0037$ | $-0.0039$ |
| | shape shared between charges | $-0.0037$ | $-0.0040$ |
| | symmetric, $A_b = 0$ | $-0.0028$ | $-0.0031$ |
| | *group uncertainty* | *0.0037* | *0.0040* |
| Signal | Crystal Ball | $-0.0002$ | $-0.0006$ |
| | $\mu, \sigma$ per charge | $-0.0000$ | $-0.0001$ |
| | *group uncertainty* | *0.0002* | *0.0006* |
| Preselection | `ProbK` $> 0.4$ | $-0.0032$ (0.0038) | $-0.0032$ (0.0037) |
| | `ProbK` $> 0.6$ | $+0.0058$ (0.0048) | $+0.0058$ (0.0050) |
| | `ProbPi` $< 0.4$ | $+0.0022$ (0.0022) | $+0.0020$ (0.0023) |
| | `ProbPi` $< 0.6$ | $-0.0004$ (0.0020) | $-0.0006$ (0.0017) |
| | *group uncertainty* | *0.0058* | *0.0058* |
| Total (quadrature) | | 0.0070 | 0.0072 |

TABLE XV. Systematic study common to the Bayesian and frequentist fits: shift $\Delta$ of $A_{\text{raw}}$ for each variant and, in parentheses and for the variants that change the sample, the expected statistical spread $\sigma_B$ of the shift. The largest contributing shift of each group is its uncertainty (italics); the groups are added in quadrature. The single exponential is rejected by the data and excluded. For the window $[5150, 5600]$ the frequentist uncertainty is scaled from the number of events, since the Hesse matrix of that fit is not accurate.

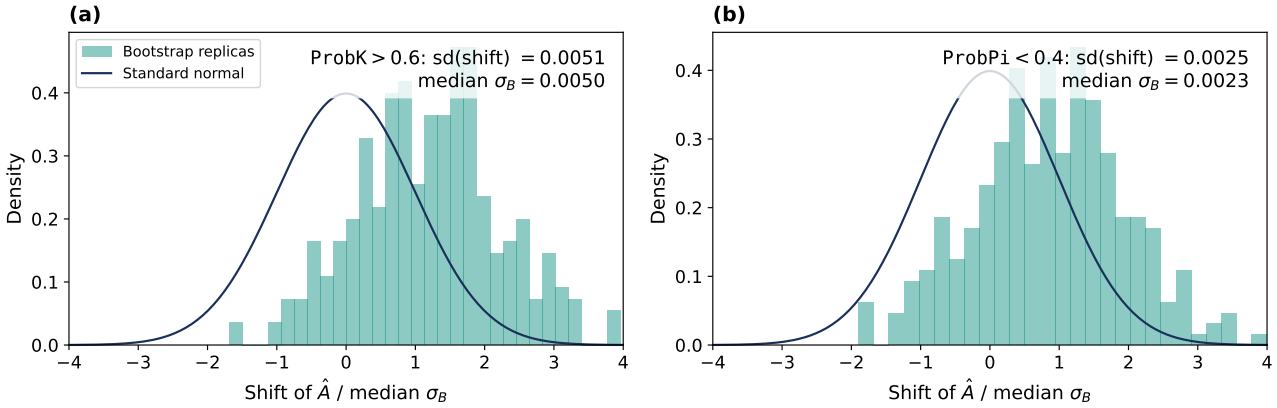



FIG. XLI. Shift of $\hat{A}$ between the nominal sample and a tighter selection, divided by the median $\sigma_B$, in 300 bootstrap replicas of the data: (a) `ProbK` $> 0.6$, (b) `ProbPi` $< 0.4$, and standard normal distribution.

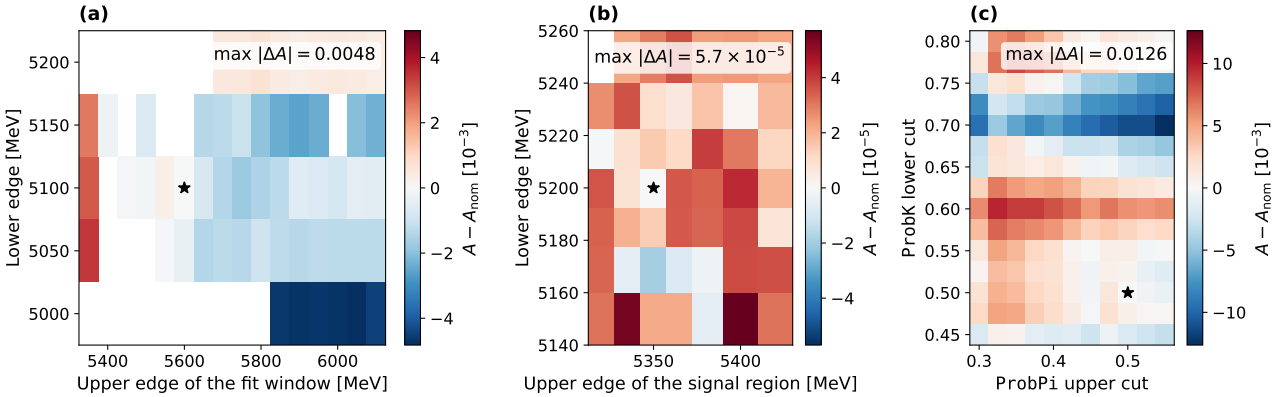



FIG. XLII. Stability of the fitted $A_{\text{raw}}$ over the three grids: (a) lower and upper edge of the fit window (only fits with a valid minimum and an accurate covariance are shown); (b) lower and upper edge of the signal region; (c) lower cut on `ProbK` and upper cut on `ProbPi`. The colour is the shift with respect to the nominal fit (star), in units of $10^{-3}$ for (a) and (c) and $10^{-5}$ for (b); the statistical uncertainty of the nominal fit is $8.5 \times 10^{-3}$.

### Appendix I: Comparison of the Two Approaches

*In short: the profile likelihood against the posterior and the two magnet polarities, the confidence and credible regions in $(A, A_b)$, and the two methods at low statistics.*

**1. Profile Likelihood, Posterior and the Two Polarities** *In short: profiling and marginalizing the nuisance parameters give the same distribution of $A$ (Fig. XIV); the two magnet polarities agree with each other, and combining their likelihoods reproduces the joint fit (Table XVI).*

Table XVI gives $A_{\text{raw}}$ for the two magnet polarities and their combination, by the two methods: the frequentist combination is the sum of the profile likelihoods, the Bayesian one the sequential update of the posterior.

**2. Confidence and Credible Regions in $(A, A_b)$** *In short: the 68% and 95% regions of the frequentist profile ($\Delta$NLL at the $\chi^2_2$ values 2.30 and 5.99) and of the Bayesian highest posterior density coincide, and $A = 0$ lies outside both (Fig. XLIII).*

Figure XLIII shows the 68% (solid) and 95% (dashed) regions in $(A, A_b)$, frequentist and Bayesian, with the best-fit point and the posterior median.

**3. Bayesian and Frequentist Intervals at Low Statistics** *In short: credible and confidence intervals coincide for $\sim 2 \times 10^4$ candidates with flat priors, and stay equal, in value and coverage, down to 50 candidates (Fig. XLIV).*

Figure XLIV shows the coverage of the 68% (a) and 95% (b) intervals on $A$ against the number of candidates, for the Bayesian credible, the Minos and the Hesse intervals, in 300 toy-MC datasets per point.

| | Frequentist | Bayesian |
|---|---|---|
| Magnet down | $-0.0492 \pm 0.0111$ | $-0.0497 \pm 0.0112$ |
| Magnet up | $-0.0374 \pm 0.0131$ | $-0.0378 \pm 0.0129$ |
| Combination of the two | $-0.0443 \pm 0.0084$ | $-0.0446 \pm 0.0086$ |
| Joint fit | $-0.0441 \pm 0.0085$ | $-0.0443 \pm 0.0085$ |

TABLE XVI. $A_{\text{raw}}$ for the two magnet polarities and their combination, by the two methods; the frequentist combination is the sum of the profile likelihoods of the two polarities, the Bayesian one the sequential update of the posterior.

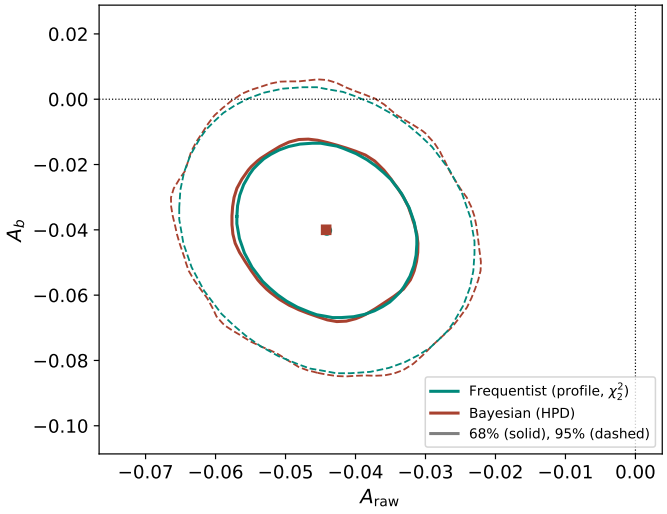



FIG. XLIII. 68% (solid) and 95% (dashed) regions in $(A, A_b)$: frequentist, from the profile likelihood with $\chi^2_2$ thresholds, and Bayesian, highest-posterior-density regions of the chain. Dots: best-fit and posterior median.

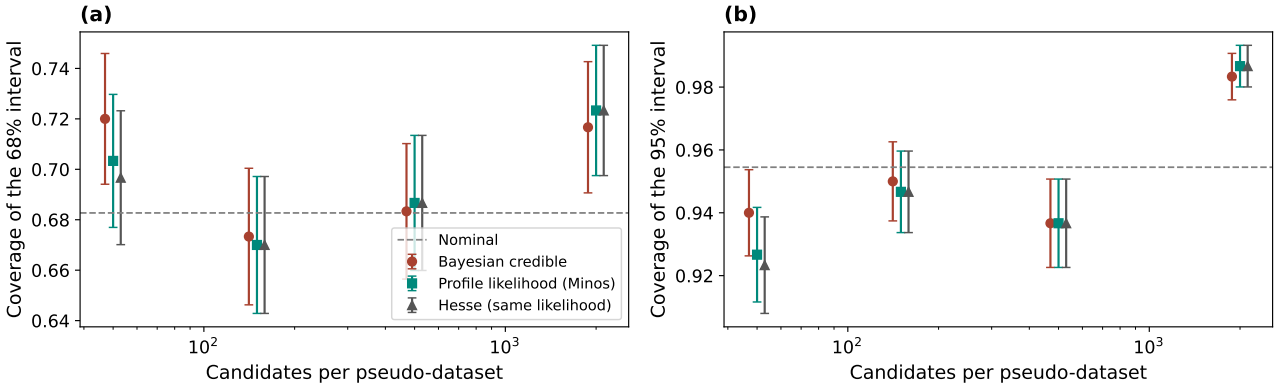



FIG. XLIV. Coverage of the (a) 68% and (b) 95% intervals on $A$ vs the number of candidates, for the Bayesian credible interval, the profile-likelihood (Minos) interval and the Hesse interval of the same likelihood; 300 toy-MC datasets per point, binomial uncertainties. The dashed lines mark the nominal coverage ($1\sigma$, $2\sigma$; the Bayesian 95% interval is the 2.5–97.5% one).

### Appendix J: Notation

*In short: the symbols and abbreviations used in the report.* The symbols used in the text are listed in Table XVII.

| Symbol | Meaning |
|---|---|
| $A_{\text{raw}}$ | raw charge asymmetry $(N^- - N^+)/(N^- + N^+)$ of the signal, parameter $A$ of the fit |
| $A_{CP}$, $A_P$, $A_D$ | CP asymmetry; production asymmetry of $B^\pm$; detection asymmetry |
| $A_{\text{count}}$ | counting asymmetry of all candidates of a sample |
| $A_b$ | asymmetry of the background yields |
| $N_s$, $N_b$ | signal and background yields (both charges) |
| $\mu$, $\sigma$ | mean and width of the Gaussian signal |
| $\lambda^\pm_{1,2}$, $f_\pm$ | slopes and fraction of the double-exponential background, per charge |
| $\theta$ | vector of the twelve parameters |
| $\mathcal{L}$, $\pi$ | likelihood and prior |
| $q(A) = 2\Delta\text{NLL}$ | profile likelihood ratio of $A$; $t(A_{\text{true}})$ at the injected value |
| $Z$, $p$ | significance and $p$-value of the test of $A = 0$ |
| $BF_{01}$, $BF_{10}$ | Bayes factor of $A = 0$ against $A$ free, and its inverse |
| $p_B$ | posterior predictive $p$-value |
| $\sigma_B$ | expected spread of the shift between nested samples, $\sqrt{\lvert\sigma^2_{\text{var}} - \sigma^2_{\text{nom}}\rvert}$ |
| $N/\tau$ | chain length in integrated autocorrelation times |
| $S_i$, $S_k$ | local significance of a Dalitz bin or cluster with respect to the global asymmetry |

TABLE XVII. Main symbols used in the text.


[1] R. Aaij *et al.* (LHCb), Direct $CP$ violation in charmless three-body decays of $B^\pm$ mesons, Phys. Rev. D **108**, 012008 (2023), arXiv:2206.07622 [hep-ex].

[2] L. Falcão, CP violation measurements in three-body charmless B decays, on behalf of the LHCb Collaboration, 12th Workshop on the CKM Unitarity Triangle (2024), arXiv:2403.19921 [hep-ex].

[3] LHCb Collaboration, Matter Antimatter Differences – B meson decays to three hadrons (2017), record 4900.

[4] LHCb Collaboration, Matter Antimatter Differences – Project Notebook (2017), record 4902.

[5] LHCb Collaboration, R. Aaij, *et al.*, Measurement of CP violation in the phase space of $B^\pm \to K^\pm\pi^+\pi^-$ and $B^\pm \to K^\pm K^+ K^-$ decays, Physical Review Letters **111**, 101801 (2013), arXiv:1306.1246.

[6] H. Dembinski, P. Ongmongkolkul, *et al.*, scikit-hep/iminuit, Zenodo (2020).

[7] F. James and M. Roos, Minuit – a system for function minimization and analysis of the parameter errors and correlations, Computer Physics Communications **10**, 343 (1975).

[8] S. S. Wilks, The large-sample distribution of the likelihood ratio for testing composite hypotheses, Annals of Mathematical Statistics **9**, 60 (1938).

[9] J. Goodman and J. Weare, Ensemble samplers with affine invariance, Commun. Appl. Math. Comput. Sci. **5**, 65 (2010).

[10] D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman, emcee: The MCMC Hammer, Publ. Astron. Soc. Pac. **125**, 306 (2013), arXiv:1202.3665.

[11] J. M. Dickey, The weighted likelihood ratio, linear hypotheses on normal location parameters, Ann. Math. Statist. **42**, 204 (1971).

[12] H. Jeffreys, *Theory of Probability*, 3rd ed. (Oxford University Press, Oxford, 1961).

[13] E. M. Metodiev, B. Nachman, and J. Thaler, Classification without labels: Learning from mixed samples in high energy physics, JHEP **10**, 174, arXiv:1708.02949.

[14] R. Barlow, Systematic errors: facts and fictions, in *Conference on Advanced Statistical Techniques in Particle Physics* (Durham, 2002) arXiv:hep-ex/0207026.